



On the development of part-scale FEM modeling for laser powder bed fusion of AISI 316L stainless steel with experimental verification

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Abstract

In laser powder bed fusion (LPBF), the effects of operating conditions on thermal gradients and residual stresses are the utmost challenges that require significant attention. The magnitudes of residual stress in the printed layers, as well as the distribution along the printed components, have not been well explained for LPBF parts. In this study, a 3D finite element thermo-mechanical model has been established to investigate the effect of operating conditions on thermal distribution, melt pool evolution, residual stress distribution, and part distortion. The printed AISI 316L stainless steel cubes have been characterized experimentally. The results showed a proportional correlation among the number of layers, thermal distribution, and melt pool dimensions. A combination of compressive and tensile stresses was recorded in the LPBF-ed parts. The Cauchy stresses were maximum in magnitude at the bottom and top surfaces along the xx- and yy-orientations, while these stresses increased in magnitude along with the part-build orientation (zz) within the whole printed cube except the top surface. The Von Mises stresses were minimal than Cauchy stresses. A maximum displacement was identified at the printed components' contours, gradually decreasing from top to side walls and top surface. An inverse correlation was identified among average Von Mises stresses (AVMS), laser power (LP), and hatch distance (HD); however, a proportional relationship is presented between laser scanning speed (LSS) and AVMS. The average displacement (AD) presented an inverse relationship with LSS and HD, while a proportional correlation has been presented between LP and AD. Average thermal distribution (ATD) revealed an inverse effect on AVMS and a proportional effect on AD. In the printed parts, only austenite-gamma phase was identified along (111), (200), and (220) orientations, with a lack-of-fusion defect in the morphology.

Keywords Laser powder bed fusion · AISI 316L stainless steel · Thermal distribution · Melt pool evolution · Phase formation · Morphology · Surface roughness · FE-modeling and experiments

1 Introduction

According to ISO/ASTM 52,900:2021, additive manufacturing (AM) refers to technologies that sequentially join material to create physical objects based on 3D model data [1]. AM processes can be classified into binder jetting, directed energy deposition, material extrusion, material jetting, powder bed fusion, sheet lamination, and vat photopolymerization [1–3]. Laser powder bed fusion (LPBF) is a metal additive manufacturing technology that can be used to produce items with high-dimensional control and complex geometries [4]. LPBF can produce components that need less assembly, reduce material waste, and contain unique internal pathways compared to conventional production [5]. The biomedical, automotive, oil and gas, industrial, and aerospace sectors have developed a strong interest in LPBF

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technologies due to the advantages mentioned above. Steel-based alloys have been the primary engineering materials for structural and sub-structural applications. They have become an integral part of our daily lives, and their significance to our civilization is shown in numerous applications. Aerospace, automotive, medical, machinery, nuclear reactors, oil and gas, food, and consumer applications are among the top industries utilizing steel. Steels provide stronger strength, higher corrosion resistance, good ductility, excellent toughness, and low cost, among other alloys. AISI 316L stainless steel (SS) is extensively used owing to the low cost, ease of processing, outstanding corrosion resistance, excellent toughness, and elevated performance under harsh working circumstances compared to the other steel grades.

In LPBF, various researches, including parameter optimization [6], mechanical property evaluation [7], and microstructure formation [8], have been carried out to improve the manufacturing process of SS 316L. However, the most prevailing defects in LPBF of SS 316L include porosity [9], lack of fusion [10], balling [11], residual stresses [12], and warpage and distortion [13]. These flaws, particularly the high residual stresses in the LPBF process, limit the load-bearing capability of the SS 316L components. One way is to use post-printing heat treatments to reduce these flaws [14]. Extremely high thermal gradients and quick melting-solidification in LPBF control all the features in the printed components. Cyclic heating and cooling are a crucial element that can greatly impact the LPBF of SS 316L, as it can modify the microstructure and mechanical characteristics of the final component [15]. LPBF includes the rapid melting and solidification of metal powder, resulting in a rapid cooling rate that can facilitate the creation of a fine-grained microstructure with increased mechanical characteristics [15]. However, repetitive heating and cooling can produce grain development and coarsening, which reduces the part's strength and toughness [15]. In addition, thermal cycling can create residual stresses, resulting in warping, cracking, or distortion, as well as thermal fatigue, which may lead to the formation of flaws or fractures [16]. Other metallic systems, such as tungsten-based alloys [17], aluminum alloys [18], composite materials [19], and copper alloys [20], can also be produced using LPBF. The difficulties encountered with these metallic systems are diverse and vary based on the chemical and physical characteristics of the feedstock. Some problems are frequent, such as residual stresses, deformation, and porosity; however, the degree of such defects and their underlying causes may differ greatly from SS 316L. Hence, the study of thermal distribution (TD), residual stress (R-S) formation, and corresponding morphology is extremely important in LPBF-ed parts. Various experimental and finite element (FE) simulation models have been applied for such measurements.

During LPBF of metallic materials, the strain evolution was identified inversely correlated with laser scanning speed [21]. For LPBF of steel, residual stress distribution was measured experimentally [22]. The largest stress distribution was identified at the deposited layer top, which was reduced up to 70% by heat treatment [22]. In another study, the influence of energy density (ED) was studied on residual stress distribution. ED played a critical role in defining the thermal distribution and final R-S such that a decrease in ED will reduce the R-S magnitude [23]. The influence of laser-related parameters on the R-S distribution was identified [24]. It was identified that the R-S magnitude dropped at the top of the printed layers when laser scanning speed decreased, or laser power increased. R-S can be controlled by reducing the layer length and laser scanning speed such that a short and fast scan will eliminate the R-S [25]. The effect of operating condition was monitored during LPBF of SS 316L, and it was identified that the tensile stresses increased along with the part thickness [26]. Besides, R-S, anisotropy and mechanical properties of printed parts can be reduced using milling [27] and post-machining [28] operations. In this context, peripheral milling operations were carried out to analyze the interactions between the machining parameters, layer thickness, and the microstructural properties of printed material [27]. It was discovered that the low-ED sample had more anisotropy in the cutting force than the high-ED sample. Post-machining process was used to improve the Young modulus and surface morphology of the Inconel 718 part [28]. It was identified that post-machining improved the Young modulus up to 10% while the surface roughness and cutting forces reduced by 50% in the LPBF-ed parts.

Besides experimental analyses, another approach is to apply the FE modeling technique to quantify the thermal distribution and residual stress distribution within the printed parts. Using the FE approach, the influence of laser power and scanning speed was investigated on the R-S distribution [29]. The stress values were determined to be maximum around the base plate (680 MPa) and reduced (350 MPa) while moving from the bottom point towards an upward direction. A 3D FE-model was developed for LPBF of SS 316L, showing that the R-S were maximum along with the laser scanning direction [30]. A relation between R-S and operating conditions was quantified via the FE model and exhibited that each condition plays a different role in the stress magnitude. The laser scanning speed influences R-S magnitude significantly compared to the laser power and hatch distance [31]. In the case of Inconel 718, the outcome of the FE model showed that the R-S increase (21.7%) when laser power elevates from 250 to 450 W, while the R-S decreases (13.6%) when laser scanning speed increases from 500 to 1000 mm/s [32]. Besides,

the R-S also increase (33.1%) when the layer thickness is enhanced from 20 to 60 μm [32].

Based on the provided literature, it can be said that most of the studies were focused on the single-layer or a few layers' depositions only. However, there is a strong need for part-scale modeling to analyze the thermal history, melt pool evolution, and stress-strain phenomena in LPBF parts as the multi-physics and heating and cooling cycles become complex. Furthermore, a few parameters, including laser power, laser scanning speed, hatch distance, and layer thickness, were considered during analyses. Recently, new LPBF machines involve various unique features such as powder bed pre-heating, different scanning strategies, and varying gas flows, which may change the whole story behind the thermal distribution and residual stress induction. Recently, the researchers and scientists have explored the impact of various scanning strategies over printed part properties. Giganto et al. [33] studied the effect of normal, concentric, and hexagonal scanning strategies on 17-4 PH porosity and mechanical properties. It was found that hexagonal scanning strategy provided least porosity, while hexagonal and normal strategies provided good mechanical characteristics and dimensional stability. In another study by Leo et al. [34], the effect of operating conditions, particularly laser beam radius and beam defocus, was studied on microstructure, defects, and heat treatment for 17-4 PH LPBF components. According to the results, extending the overlap from 25 to 50 μm and decreasing the laser beam scan radius from 5000 to 1250 μm both led to an increase in the austenitic phase relative to an in-focus treated sample. The effect of gas and its flow on the LPBF-ed parts' characteristics is a significant aspect that must be carefully considered [35]. Inert gases, such as argon and nitrogen, are used to eliminate the possibility of oxidation during the printing process, ensuring the manufacture of high-quality parts [36]. Furthermore, the heat transfer and thermal distribution within the printing chamber are strongly controlled by gas flow, which is critical in manufacturing parts with homogeneous characteristics [37]. A turbulent gas flow can distort the final part, whereas insufficient gas flow rates might lead to a drop in LPBF-ed components density, thus impacting component porosity and mechanical qualities [38].

The residual stress magnitudes in the printed layers and residual stress distribution along the printed components have not been well-explained in location and various orientations of LPBF parts. Hence, this area requires attention from researchers and scientists. By considering the gap explained above, a 3D FE thermo-mechanical model has been proposed containing laser power, laser scanning speed, hatch distance, powder layer thickness, powder bed pre-heating, and laser scanning strategy. This framework was applied to analyze the effect of the number of layers on thermal distribution, melt pool evolution, residual stress distribution, and

part distortion. The LPBF-ed SS 316L cubes ($10 \times 10 \times 10 \text{ mm}^3$) were subjected to scanning electron microscopy, X-ray diffraction, and profilometric analyses to correlate operating conditions and morphology, phase formation, and roughness parameters. This study will help the LPBF community to understand the thermal distribution, melt pool evolution, residual stress distribution, part distortion, phase transformation, and roughness parameters as a function of operating conditions during the printing of SS 316L.

2 Methodology proposed

This section has been classified into finite element (FE) modeling, LPBF experiments, and characterization of the LPBF parts.

2.1 Thermo-mechanical analysis modeling

The governing equation for the 3D transient thermal analyses can be expressed as [39, 40]

$$\rho C_p \frac{\partial T}{\partial t} = \nabla(k \nabla T) + Q_{\text{rad}}. \quad (1)$$

Here, ρ , C_p , t , and k are the material's density, material's specific heat, time, and material's thermal conductivity, respectively. Here, Q_{rad} (W/mm^3) is the moving heat source with the Gaussian energy distribution, expressed as [39, 41, 42]

$$Q_{\text{rad}} = \frac{2P}{\pi r_o^2 H} \left(1 - \frac{h}{H}\right) e^{\left(1 - \left(\frac{r}{r_o}\right)^2\right)}, \quad (2)$$

where P , r_o , H , r , and h are laser power, beam radius, beam penetration depth, distance from the beam axis where the intensity becomes $1/e^2$ ($\approx 13.5\%$) of the highest value, and convection coefficient, respectively. The material's cooling at the build surface was considered by radiation and convection as [39, 41, 42]

$$q_{\text{rad}} = \sigma \epsilon (T_o^4 - T_{\infty}^4). \quad (3)$$

$$q_{\text{conv}} = h_{\text{conv}} (T_o - T_{\infty}). \quad (4)$$

Here, ϵ , σ , T_o , and h_{conv} are the materials' surface emissivity, Stefan-Boltzmann constant, surface temperature during printing, and connective heat transfer, respectively. The thermally induced distortion in the AM parts by strain parameters [39, 41, 42] is as follows:

$$\epsilon^* = H^{1.5} \frac{\delta \Delta T_{\text{am}}}{EF \sqrt{\rho}}. \quad (5)$$

Equation 5 contains δ , ΔT , t_{am} , H , E , and F designated as coefficient of expansion, thermal distribution difference between current and room temperatures, time required to manufacture components, flexural rigidity of a given material, and Fourier transformation measuring the heat dissipation. A quasi-static mechanical analysis was carried out to examine the stresses (σ) and deformation formation as [39, 41, 42]

$$\nabla \sigma = 0. \quad (6)$$

$$\sigma = C \varepsilon_e. \quad (7)$$

$$\varepsilon = \varepsilon_e + \varepsilon_p + \varepsilon_T. \quad (8)$$

Here, C is the material stiffness with fourth order, ε is the total strain, ε_e is the elastic strain, ε_p is the plastic strain, and ε_T is the thermal strain. Table 1 collects the thermo-physical characteristics of AISI 316L stainless steel (SS) used in the simulation model. All the simulations were carried out in ANSYS Additive, USA (2019 R3 Version).

2.2 LPBF experiments

The gas atomized SS 316L powder particulates were used to manufacture LPBF samples using ENAVISION 250, ERMAKSAN, Turkey, LPBF machine. ENAVISION 250 has been designed and developed at ERMAKSAN, Turkey, laser source similar to Trump while EOS and Renishaw use laser source developed from IPG Photonics. The particle size distribution (PSD) is $D_{90}=46 \mu\text{m}$, $D_{50}=30 \mu\text{m}$, and $D_{10}=19 \mu\text{m}$.

Table 1 SS316L thermo-physical characteristics [40]

Quantity	Value
Liquid temperature	1673 K
Solid temperature	1643 K
Thermal conductivity	$11.82 + 0.011236 T$
Specific heat	$330.9 + 0.563 T - 4.015 \times 10^{-4} T^2 + 9.465 \times 10^{-8} T^3$
Density	7990 kgm^{-3}
Latent heat of fusion	$272 \times 10^3 \text{ Jkg}^{-1}$
Liquid laser absorptivity	70%
Powder bed laser absorptivity	30%
Thermal expansion	$16.5 \times 10^{-6} \text{ kg}^{-1}$
Young's modulus	185 GPa
Viscosity	$7 \times 10^{-3} \text{ kgms}^{-1}$

Table 2 SS 316L chemical composition [40]

Elements	Fe	Cr	Ni	Mo	Mn	Si	C	N	P	S
Wt. (%)	Bal	16.0–18.0	10.0–14.0	2.0–3.0	2.0	0.750	0.030	0.100	0.045	0.030

This machine is equipped with a fiber laser double scanning head delivering a maximum Gaussian laser power and scanning speed up to 500 W and 11 m/s, respectively, along with the 392 °F substrate heating and $250 \times 250 \times 300 \text{ mm}^3$ build volume. In the current study, SS 316L powder particulates were printed on a SS 316L base plate under inert atmosphere provided by argon gas. The chemical composition of SS 316L feedstock is provided in Table 2.

In Table 3, single-layer experiments were carried out to validate the FEM model. The parameters, including laser power = 150 W, powder layer thickness = 30 μm , and scanning speed = 700 mm/s, have been recommended by the ERMAKSAN, Turkey, for SS316L. The sensitivity analysis has been carried out by changing the laser power for FEM model validation and robustness at various laser energy densities, as provided in Fig. 2. After FEM model validation, the endorsed FEM model was converted from single layer to multiple layers. For experimental analyses, experiments were designed using full factorial method by using laser power = 100–250 W (interval of 10W), laser scanning speed = 500–1500 mm/s (interval of 100 mm/s), hatch distance = 100–150 μm (interval of 10 μm), and powder layer thickness = 30, 50, and 70 μm . These set of parameters provided a range of operating conditions for SS316L; however, all the parameters did not yield printable samples. Here, printable samples were defined as the samples produced without considerable defects. Table 3 collects the operating conditions, for multiple-track experiments, resulted for printable samples only. Furthermore, for LPBF studies, a powder layer thickness of 30 μm was chosen. A powder layer thickness of 30 μm is commonly regarded as optimal in LPBF since it provides for a balance of build speed and component quality. When the powder layer thickness is too thin, the build speed slows, resulting in a longer time to print a part. If the powder layer thickness is too thick, the heat generated during the laser melting process will not be sufficient to completely melt the material, resulting in defects and reduced part quality. As a result, ERMAKSAN, Turkey, has advised a powder layer thickness of 30 μm to offer a suitable balance between build speed and component quality in the instance of SS 316L.

To print samples A1–A9, 67°–67° was used. In LPBF, the 67°–67° laser beam scanning approach is frequently selected over other scanning systems since it has various advantages [43, 44]:

- The 67°–67° scanning approach includes switching the laser beam direction at each layer, which aids in the equal

Table 3 Operating conditions for SS 316L experiments

Single-track experiments				
Specimen No	Laser power (W)	Powder layer thickness (μm)		Scanning speed (mm)
S1	70	30		700
S2	150	30		700
S3	190	30		700
S4	270	30		700
Multiple-track experiments (10 mm×10 mm×10 mm cubes)				
Specimen No	Laser power (W)	Laser scanning speed (mm/s)	Hatch distance (μm)	Powder layer thickness (μm)
A1	190	1400	100	30
A2	190	1200	100	30
A3	190	1000	100	30
A4	190	800	100	30
A5	190	1400	120	30
A6	200	1400	100	30
A7	210	1400	100	30
A8	220	1400	100	30
A9	230	1400	100	30

distribution of heat and the reduction of thermal stress. This can lead to less distortion and improved overall component quality.

- The laser beam overlaps the preceding layer by roughly 30% when utilizing this scanning approach, resulting in a greater surface quality and smoother finish of the final product.
- As the alternating beam direction minimizes the time necessary for each layer, the 67°-67° scanning method allows for quicker production rates while still retaining high-quality products.
- This scanning method reduces waste by preventing the production of overhanging structures, which can be difficult to sustain and need the use of additional material for support structures.

It is the reason 67°-67° scanning strategy was used in current study.

2.3 Characterization of the printed samples

After LPBF single- and multi-layer experiments, scanning electron microscopy (SEM) from Thermo-fisher Scientific (Apreo S), USA, was used to capture the printed samples' morphologies. For multi-layer analyses, $\times 260$, $\times 500$, $\times 1000$, and $\times 4000$ magnifications were applied with a combination of accelerating voltage = 10.00 kV and current = 0.10 nA in T2 detector mode. It is important to mention here that SEM

was performed at the top of the as-built specimen perpendicular to the build-orientation. The samples were tested under high voltage with the Optiplan mode. Optiplan mode is suitable for imaging thick and three-dimensional SS 316L samples. Optiplan mode can also be useful for obtaining a clearer image of surface features or defects in the SS 316L samples. To capture the cross-sections of the printed single layers, the cross-sections were prepared using planar grinding (SiC with water), pre-polishing (alpha with 8- μm polishing solution), polishing (gamma with 3- μm polishing solution), final polishing (lambda with 0.06- μm polishing solution), and etching (V2A). To identify the effect of operating conditions over crystallographic structure of SS 316L, X-ray diffraction (XRD) was conducted via D8 Advance enclosure from Bruker, USA. Table 4 collects the operating conditions applied for XRD analyses of SS 316L.

All the LPBF-ed cubes were subjected to surface roughness analyses using profilometer from Mitutoyo SurfTest, SJ-201, Japan, to determine the effect of operating conditions, including laser power, laser scanning speed, and hatch distance. During surface roughness testing, sampling length = 0.8 mm and number of sampling spans = X5 were chosen which resulted in Ra, Rz, and Rq values. It is important to mention here that, in the provided region, three readings were taken, and an average value was calculated for a better statistical analyses of roughness-related parameters. According to ISO 4287-1:1984, Ra is the average surface roughness, defined as the arithmetic average of the profile

Table 4 Operating condition XRD analyses of SS 316L

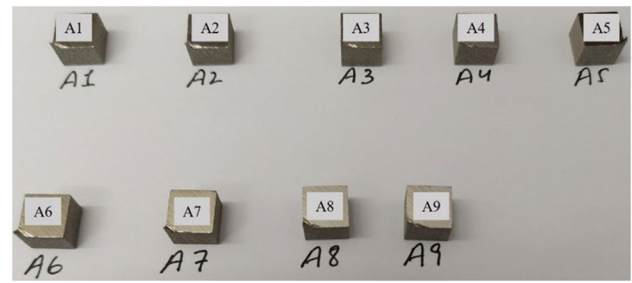
Item	Value
Scan range	20–80°
Scan speed	1.00°/min
Scan axis	2 theta/theta
Scan mode	Continuous

height absolute values over the sampling length [45]. R_z , designated as average ten-point height, is an average value of five highest-profile peaks and five lowest valleys within the sampling length [45]. Root mean square average of profile heights across an evaluation length defines R_q , also commonly known as RMS roughness [45]. It is essential to mention here that the roughness parameters were calculated along (x-direction) and perpendicular (y-direction) to the powder spreader movement. From literature studies [46, 47], it has been identified that the surface roughness parameters vary in both directions due to the morphological results imposed during powder spreading and corresponding melting.

For simulation, ANSYS software carries out geometric dimensioning and tolerancing (GD&T) measurements using simulation output. In this case, the software compares the input STL file and the simulated results to estimate the GD&T values. For experimental validation, an in-house developed coordinate measuring machine (CMM) facility was applied to measure the dimensions of the printed part. The specifications of CMM include magnification = $\times 10X$ – $\times 100$, resolution = 0.001 mm at linear scale, least count value = 1.0×10^{-4} mm, contour light = Green LED, surface light = white LED, probe travel = $400 \times 380 \times 150$ mm³, and vision travel = $400 \times 430 \times 150$ mm³. The magnitude of achieved part dimensions, including length, width, and height, was measured at 15 different locations for each quantity. An average value was calculated for all the measured dimensions along the length (CL), width (CW), and height (CH) of the printed cube. These average values were subtracted from the theoretical values of length (TL), width (TW), and height (TH) attained from the CAD model. Finally, an overall experimental average value of part displacement (EAPD) was estimated by applying root-mean square formula as

$$\text{EAPD} = \sqrt{|\text{CL} - \text{TL}|^2 + |\text{CW} - \text{TW}|^2 + |\text{CH} - \text{TH}|^2} \quad (9)$$

It is also worth noting that the standard deviation of the measured was also calculated. However, the average values are reported here owing to the low standard deviation values. For porosity estimation, “ImageJ” was applied to measure the porosity (%) in a given area. To measure hardness of the printed cubes, Rockwell hardness testing was carried out. For this purpose, an analog Rockwell hardness tester

**Fig. 1** SS 316L samples printed by LPBF by varying laser scanning speed, hatch distance, and laser power

(MS 201-R) was utilized [48]. A ball indenter with 1/16" diameter having 10-kgf pre-load and 30-kgf testing load was applied to the printed cubes. The data was recorded for each specimen.

3 Results and discussion

In this section, the results have been classified into simulation model valuation, experimental analyses, and a correlation between simulation and experimental outputs.

3.1 Simulation model validation

Figure 1 shows the LPBF-ed SS 316L cubes, utilized for characterization. The samples A1–A4 have been printed by varying laser scanning speed, A1 and A5 samples were printed by changing hatch distance, and A6–A9 samples have been manufactured by changing the laser power, respectively.

A comparison was made between finite element (FE) simulations and experimental analyses to identify the FE model trustworthiness, as shown in Fig. 2a and b, in the case of the single printed layer's (S1–S4 in Table 3) width and depth, respectively. For this purpose, all the cross-sections were prepared using grinding and polishing and subjected to SEM analyses. One can observe that the outcome printed by FE analyses was found in close correspondence to the experimental results with decent accuracy, exhibiting the reliability of the developed FE model. During the LPBF of single-track SS 316L, Fig. 2c depicts the evolution of the melt pool and temperature distribution at the beginning, middle, and end of the powder layer, respectively. The isotherm curves indicate the SS 316L melting temperature. The highest temperature indicated by the isotherm represents the melting of the material owing to its temperature up to its melting point, resulting in an asymmetrical pool of molten material represented by a sequence of ellipses. The energy conveyed by the beam into the material was transferred together with the material's depth and rearward,

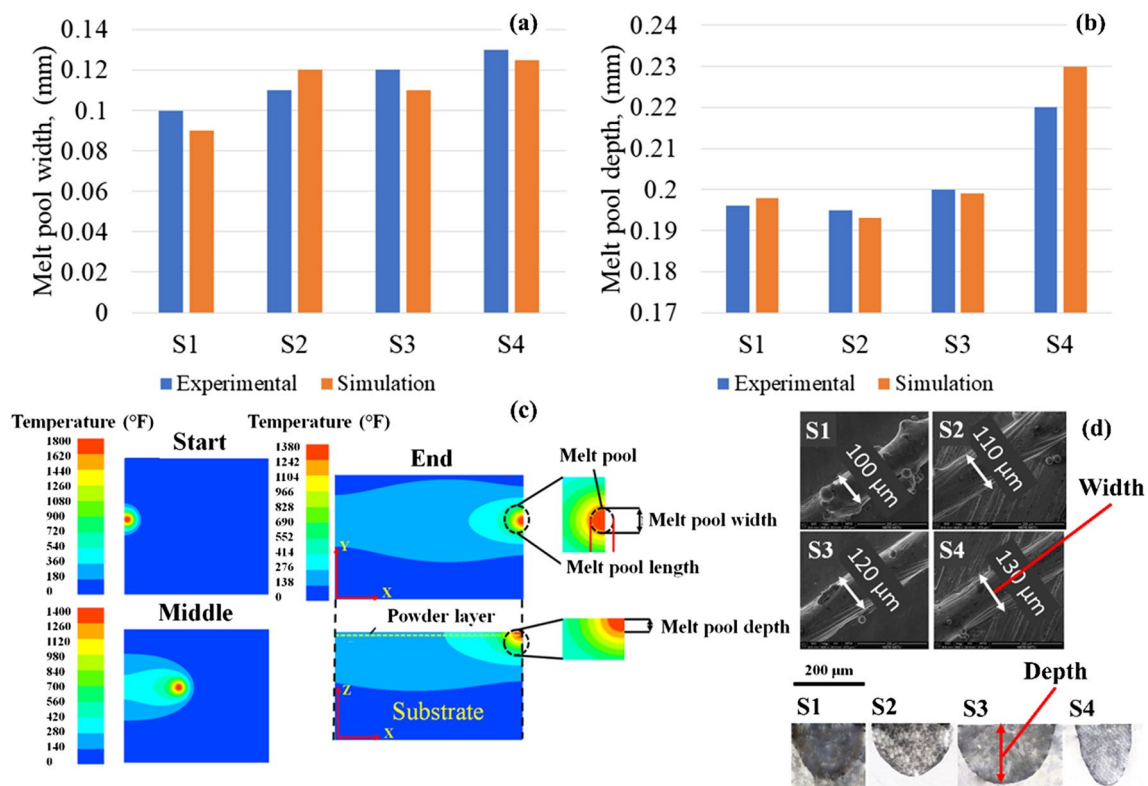


Fig. 2 Comparison between experiments and simulations for single layer by LPBF in the case of melt pool a width, and b depth, c simulation results for S1, and d SEM images for S1–S4 samples

resulting in the formation of a greater number of isotherms in the beam's front than in its rear. When the laser spot is located at the beginning of the scan, the powder layer's peak temperature reaches up to 1800 °F, exceeding the SS 316L melting temperature (1400 °F) due to the material's reduced thermal conductivity in the laser irradiation region, resulting in circumscribed melting of the powder layer. As the laser beam began to move, however, the peak thermal distribution decreased to 1400 °F because of heat conduction from the powder layer into the base plate and remained stable until the completion of the laser scan. The heat distribution throughout the powder layer and substrate after the conclusion of the laser scan confirms the declination of thermal distribution from 1800 to 1400 °F as previously described. Figure 2d shows the SEM images of the S1–S4 single-track samples compiled using 200-μm scale.

In the case of residual stress distribution within printed layers by LPBF, the results attained by current ANSYS Additive software were compared with the one achieved in Ref. [49]. In the study by Waqar et al. [49], four SS 316L layers with the dimensions equal to $1 \times 1 \times 0.12 \text{ mm}^3$ ($L \times W \times H$) were deposited over SS 316 base plate having $1.5 \times 1.5 \times 0.5 \text{ mm}^3$ geometrical dimensions

($L \times W \times H$). During LPBF simulation analyses, laser power = 200 W, hatch distance = 50 μm, and laser scanning speed = 600 mm/s were applied throughout the deposition process. The printing was carried out using 3D Systems ProX DMP 320 LPBF printing while the residual stresses were measured using a Stresstech Xstress 3000 X-ray diffractometer [49]. For current model validation, same conditions were applied as Waqar et al. [49]. In Fig. 3, it can be analyzed that the results can be classified into the minimum and maximum values of Von Mises stresses, defining a range of stresses within the deposited layers. It is essential to mention here that Waqar et al. [49] used the ABAQUS finite element (FE) analysis software, while the current study utilizes ANSYS Additive software to simulate the residual stress distribution in the printed layers. The minimum and maximum values estimated by the current simulation analysis were equivalent to 24 MPa and 176 MPa, respectively, which are pretty close to the one presented by Waqar et al. [49] 25 MPa (minimum) and 175 MPa (maximum), exhibiting the reliability of current developed framework. The difference between the two may be attributed to the thermal gradient and heating effects resulted during LPBF process.

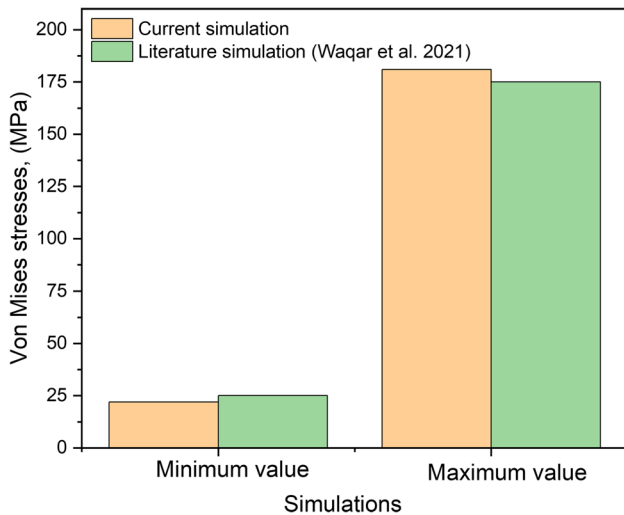


Fig. 3 Comparison between current and literature-based [49] simulations for stress formation in the printed part

3.2 Simulation analyses

After the finite element (FE) model validation, simulations were conducted for the A1–A9 cubes (see Table 3); however, the results have been presented for A1 cube only in Fig. 4. A color coding has been applied to identify the evolution of thermal distribution, and melt pool depth, length, and width, such that a yellow color, show the “highest” value while the dark red color illustrates the “least” value for a given quantity. LPBF process involves the cyclic heating and cooling for a given material along with the rapid heating with slow conduction [50], which promotes the thermal distribution elevation from 1300 to 3200 °F and the corresponding

melt pool dimensions such that depth = 0.077–0.081 mm, length = 0.08–0.74 mm, and width = 0.078–0.094 mm. The melt pool dimensions and thermal distribution are the output of ANSYS software. It is important to mention here that the LPBF involves the rapid heating and slow conduction, resulting in heat accumulation during printing. It, in return, increases the local temperature for a given material, crossing material melting, boiling, and vaporization temperatures. In LPBF, the boiling of material causes defects in printed parts. The excessive heating in the LPBF can result in printed part distortion as well as residual stresses and requires post-processing to eliminate or lessen the defects in the printed parts [51, 52]. Furthermore, the thermal distribution and melt pool dimensions can be used to estimate mechanical properties, including hardness, tensile strength, and fatigue, of the printed part [53]. However, experimental evaluation will be required to validate the results.

Figure 5a–d compiles the evolution of thermal distribution and melt pool at the 1st, 15th, 30th, and all layers during LPBF of SS 316L. It can be analyzed that the thermal distribution and melt pool dimensions, including depth, length, and width, were identified in a minimum–maximum range at each interval. The LPBF process inherits the rapid heating and melting of a given material in a localized region, followed by the material’s cool-down when the laser beam moves away from the irradiated area, resulting in solidification. In Fig. 5a, the bottom left-hand side corner has cooled down first compared to the rest of the portion, as shown by the light blue color code, while the thermal distribution and corresponding melt pool dimensions increased gradually towards the top right-hand side corner. It is because the printing starts from the bottom left-hand side and steadily moves towards the top right-hand side, resulting in a thermal

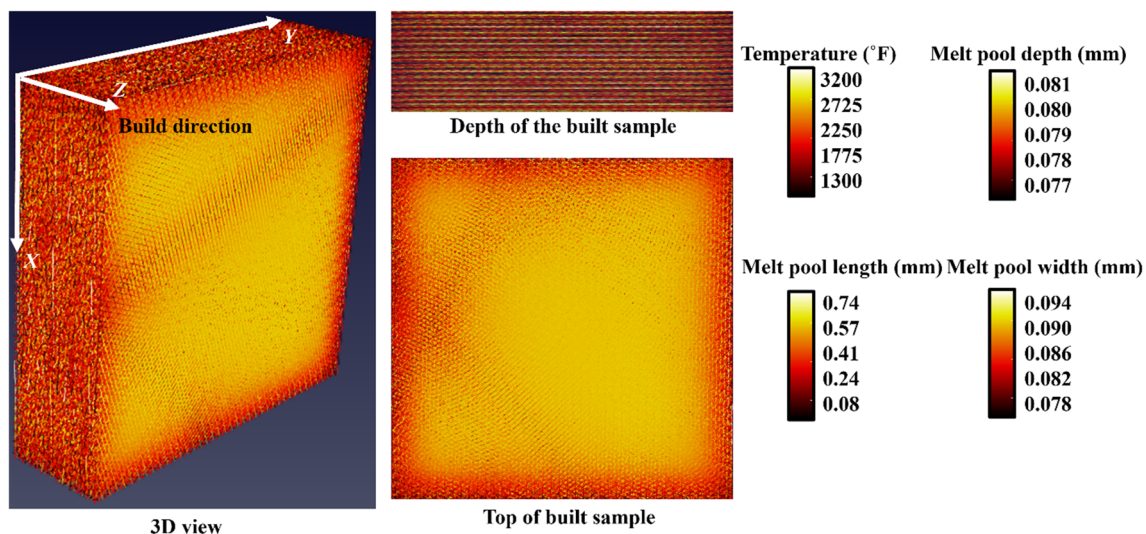


Fig. 4 Simulation analyses for thermal distribution and melt pool evolution during the LPBF of A1 sample

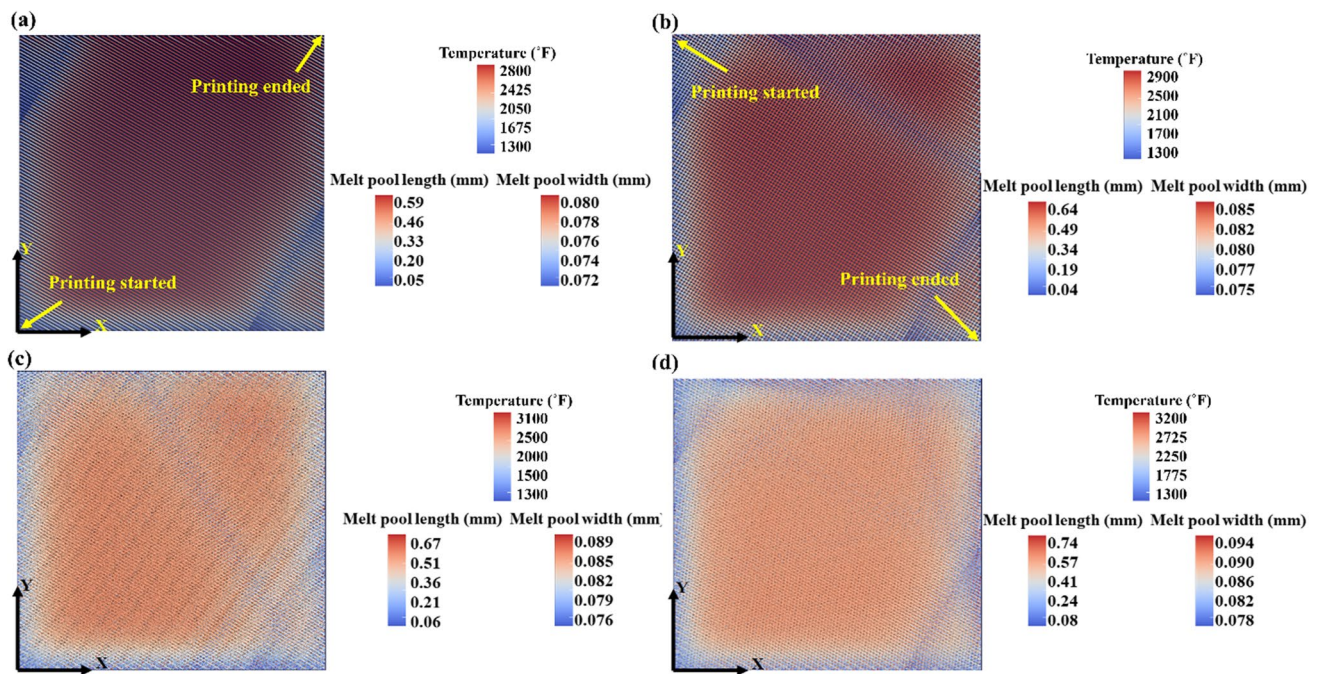


Fig. 5 Thermal distribution and melt pool evolution during LPBF of SS 316L as a function of layers' number for A1 sample in the case of 67°-67° laser beam scanning strategy: a 1st layer, b 15th layer, c 30th layer, and d all layers

distribution and melt pool dimensions variation in the whole printed region. It is important to mention that the 67°-67° laser beam scanning strategy was applied during printing. In this strategy, the subsequent layer was deposited with a 67° scanning strategy at 90° rotation of the laser beam scanning head, followed by the deposition of the previous layer to mitigate the thermal stresses and defects induced in continuous laser-material interaction. Figure 5b shows a 90° rotation of the printing start point compared to Fig. 5a. As the number of layers increases from 1 to 15, the peak values of thermal distribution and melt pool dimensions increase linearly due to the heat accumulation during the LPBF of SS 316L. The effect of surrounding powder material on the deposited layers' edges can be clearly identified in Fig. 5c, as the light blue region becomes prominent when the number of layers increases up to 30 compared to Fig. 5a. Besides, Fig. 5a presented the same trend for the thermal distribution and melt pool evolution as Fig. 5A-c. When all the layers in the 10-mm (thick) cube have been printed by LPBF, the resultant thermal distribution and melt pool dimensions can be analyzed in Fig. 5d. A proportional correlation can be identified between a thermal distribution and melt pool dimensions from the FE results. It can be explained by the fact that when the laser-material interaction time increases, more energy density is being transferred to the given area resulting in elevated melt pool dimensions. In LPBF, laser-material interaction time enhances with the number of layers keeping the operating conditions constant, which in return

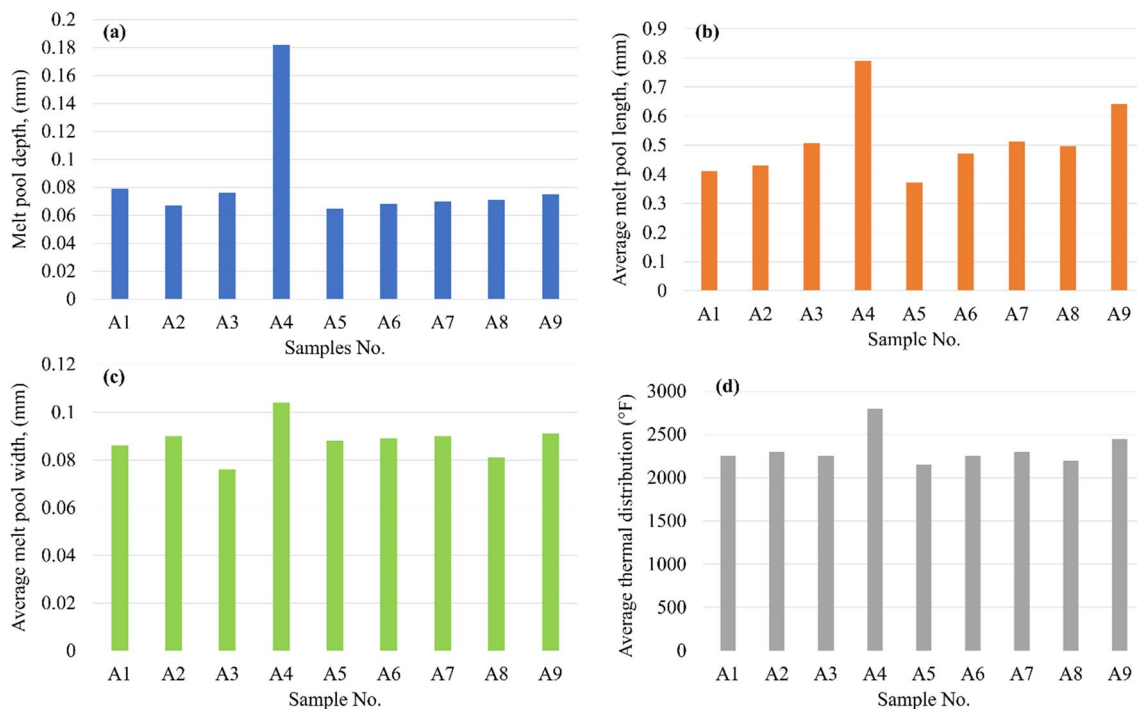
continuously increases the melt pool dimensions (depth, length, and width) and may result in lack-of-fusion, balling, and keyhole-based defects if the melt pool dimensions cross the threshold value [54].

After carrying out the finite element (FE) analyses for all the cubes (A1–A9), the melt pool dimensions, including depth, length, width, and thermal distribution, were recorded as displayed in Table 5. The dataset shows the extreme minimum and maximum values for the achieved outputs in the case of a given quantity, which were utilized to calculate average values. It is important to mention here that FEM technique cannot measure pores; however, it can be calculated by linking user-defined sub-routines with the FEM. The output of the FEM such as thermal distribution and melt pool dimensions can be used to estimate the defect [54]. The estimation accuracy depends on the precession on FEM results. In one of the recent studies [55], a user-defined routine was developed to detect pores and lack-of-fusion defects derived from reliability function evaluations and overhaul inspection findings.

Figure 6a–d compiles the average melt pool dimensions, including depth, length, width, and thermal distribution for samples A1–A9. These results have been deduced from Table 5. From the results, it can be analyzed that for A1–A4, with the decrease in laser scanning speed, an increase in average melt pool depth, length, width, and thermal distribution can be observed. In the case of samples A1 and A5, with an increase in hatch distance, the average melt pool

Table 5 Melt pool dimensions and thermal distribution evolutions during LPBF of SS 316L

Sample No	Melt pool depth (mm) (Min.–Max.)	Average melt pool depth (mm)	Melt pool length (mm) (Min.–Max.)	Average melt pool length (mm)	Melt pool width (mm) (Min.–Max.)	Average melt pool width (mm)	Thermal distribution (°F) (Min.–Max.)	Average thermal distribution (°F)
A1	0.077–0.081	0.079	0.080–0.74	0.410	0.078–0.094	0.086	1300–3200	2250
A2	0.058–0.077	0.067	0.083–0.78	0.430	0.082–0.099	0.090	1300–3300	2300
A3	0.065–0.087	0.076	0.082–0.93	0.506	0.061–0.091	0.076	1200–3300	2250
A4	0.072–0.110	0.182	0.079–1.50	0.789	0.078–0.130	0.104	1200–4400	2800
A5	0.057–0.074	0.065	0.083–0.66	0.371	0.082–0.095	0.088	1300–3000	2150
A6	0.059–0.077	0.068	0.083–0.86	0.471	0.082–0.097	0.089	1300–3200	2250
A7	0.062–0.079	0.070	0.084–0.94	0.512	0.083–0.097	0.090	1300–3300	2300
A8	0.064–0.079	0.071	0.084–0.91	0.497	0.069–0.094	0.081	1300–3100	2200
A9	0.066–0.084	0.075	0.084–1.20	0.642	0.083–0.100	0.091	1300–3600	2450

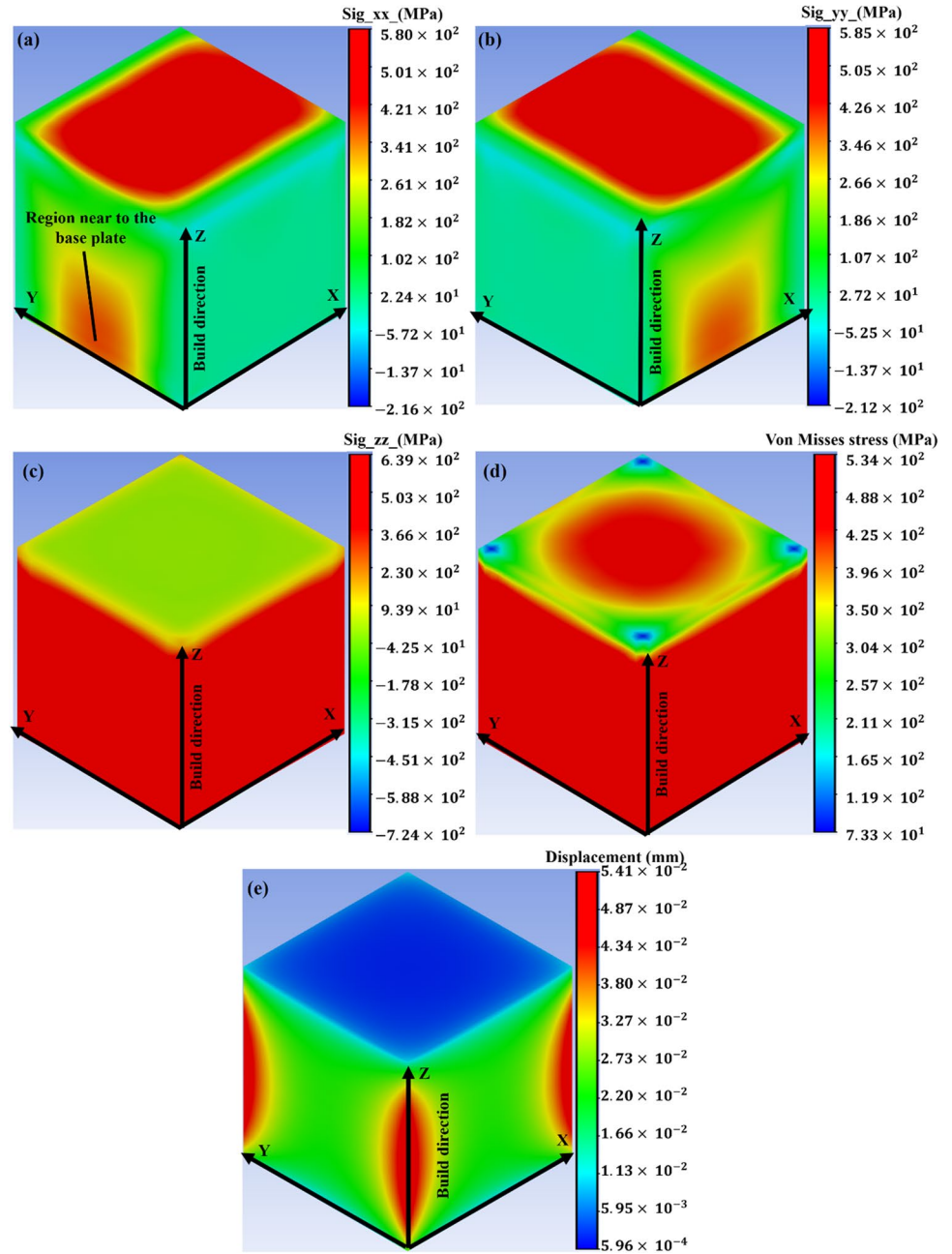
**Fig. 6** Effect of operating conditions on average melt pool a depth, b length, c width, and d thermal distribution

dimensions and thermal distribution decrease. For laser power A1 and A6–A9, with an increase in laser power, a minor increase can be observed in melt pool depth, width, and average thermal distribution. However, the average melt pool length was impacted significantly with an increment in laser power.

In LPBF, the complex transient thermal gradient involved is the most significant contributor to the development of substantial residual stresses within the printed. Thermal stresses can be used to describe residual stresses identified in LPBF-fabricated components. The stress is initially zero when the powder particles are neither heated nor melted at

a particular place. When a laser melts a point, it transforms it from powder to liquid. The liquified material starts to solidify when the laser beam moves away from the irradiated area, leading to residual stress formation. Figure 7a–e shows the Cauchy (Kirchhoff) stress distribution along xx-, yy-, and zz-orientations, Von Mises stress distribution, and displacement (distortion) within the printed SS 316L cube. All the cubes from A1–A9 were simulated using the operating conditions provided in Table 3; however, only cube A1 results are shown here. The Cauchy stress distribution is usually associated with the spatial coordinate system and explains the stress magnitude in the given orientation [56].

Fig. 7 **a** xx -orientation-, **b** yy -orientation-, and **c** zz -orientation-based-Cauchy stresses; **d** Von Mises stresses; and **e** displacement in the A1 SS 316L cube by LPBF



Von Mises stresses help in identifying that the given material will yield or rupture. In other words, these stresses help in identifying the extreme distortion strain energy criterion, suggesting that yielding starts when the elastic energy of distortion arrives at the critical value [57].

As shown in Fig. 7a, the maximum stress distribution can be identified in the region near the base plate for the xx -stress component. Note that a powder-bed pre-heating equal to the 248 °F was applied during experiments and simulations. Powder bed pre-heating plays an essential role in reducing the thermal stresses induced during laser-material interaction; however, the pre-heating should be enough to

reduce the thermal stresses in the printed components without introducing the keyhole defects. In the current scenario, the provided pre-heating was not enough to eliminate the residual stresses near the base plate, which decreased as the number of layers increases while moving from the region near the base plate to the printed cube's center. Besides, the cube was fixed near the base plate, which induces the "constrained" nature between the part and base plate, resulting maximum stresses at this location and the material expansion and contraction upon heating and cooling. Furthermore, the maximum stress was identified at the top of the printed layer due to the heating cycles involved during the LPBF of

SS 316L. The same trend can be identified along with the yy-stress component, as shown in Fig. 7b. The stress components along the xx- and yy-directions are equivalent with a minor deviation, showing a combination of compressive (negative sign) and tensile (positive sign) stresses during the LPBF process. In laser-material interaction, two phenomena are responsible for compressive and tensile residual stress formation within the printed part: “thermal gradient and cool-down.” The first category occurs when the given material experiences rapid heating and slow conduction, resulting in a high thermal gradient. It, in return, decreases the given material’s strength, which causes the material’s extension; however, the material beneath it does not allow an extension, promoting “compressive” stresses. On the other hand, when the laser beam moves away from the irradiated regime, the material starts to cool down and is counteracted by the material beneath, resulting “tensile” stress formation.

Figure 7c shows the stress distribution along with the zz-orientation, also designated as the part-build direction. One should notice that the tensile and compressive stresses are in higher magnitude along the zz-orientation compared to the xx- and yy-orientations. Since the zz-orientation is the part-build direction, the material experiences several heating and cooling cycles, promoting the material’s re-melting and solidification, thus inducing stresses and build-up. Besides, while moving from the bottom towards the upward direction along the zz-orientation, the stresses have resulted as the printed part is surrounded by the powder material in combination with the cyclic heat transfer from the top side towards the bottom end and with the surrounding environment, thus promoting compressive-tensile stresses. Hence, the stress distribution is minimum at the top of the printed specimen compared to the rest of the components. Figure 7d exhibits the Von Mises stress distribution within the LPBF-ed component, lower in magnitude than the Cauchy stresses along the xx-, yy-, and zz-orientations. The maximum stress distribution was recorded along the part-build direction due to cyclic heating and cooling cycles involved along with

this orientation. These stresses play a critical role in determining whether the provided material will yield or rupture; these stresses assist in recognizing the extreme distortion strain energy criterion. Due to the importance and criticality, such stresses have been adopted for further analyses. The displacement/distortion in the LPBF-ed component has been presented in Fig. 7e. The contours of the printed cube experienced the maximum distortion due to material tension–compression resulting from an elevated heat transfer area and the corresponding rate provided by the ambient conditions and powder material surrounding the printed cube. This distortion gradually decreases towards the side-walls and the top of the printed cube.

After carrying the FE simulation and experimental analyses for all the cubes (A1–A9), the results have been compiled in Table 6. A range defining the minimum–maximum values of Von Mises stresses and displacement has been summarized here. The simulation average values have been calculated based on the minimum–maximum values, while the average experimental displacement values have been on the criteria mentioned in Sect. 2.3.

The relationship among average Von Mises stress (AVMS), average displacement (AD), and laser scanning speed (LSS) at laser power = 190 W, hatch distance = 100 μm , and powder layer thickness = 30 μm is presented in Fig. 8a. The trends show that AVMS increased with increasing LSS, while an opposite relationship has been recorded between AD and LSS. Increasing the LSS from 800 to 1000 mm/s leads to an increase in AVMS from 285 to 297.5 MPa. The further increment in LSS from 1000 to 1200 and 1400 mm/s leads to an increase of 1.5 MPa and 4.5 MPa, respectively, with reference to the 297.5 MPa; however, this increment is not consistent. For AD, 0.045-mm, 0.023-mm, 0.03-mm, and 0.027-mm distortion values were attained for 800, 1000, 1200, and 1400 mm/s LSS. The effect of hatch distance on the AVMS and AD can be analyzed in Fig. 8b. When the hatch distance (HD) increases from 100 to 120 μm , the AVMS and AD decrease from 303.5 to 300.5 MPa and 0.027 to 0.021 mm, respectively. The

Table 6 Von Mises stresses and displacement in the LPBF-ed SS 316L cubes

Sample No	Von Mises stresses (MPa) (Min.–Max.)	Average Von Mises stresses (MPa)	Simulation displacement (mm) (Min.–Max.)	Average simulation displacement (mm)	Average experimental displacement (mm)
A1	7.33×10^1 – 5.34×10^2	303.65	5.96×10^{-4} – 5.41×10^{-2}	0.027	0.030
A2	6.47×10^1 – 5.35×10^2	299.85	5.14×10^{-4} – 5.92×10^{-2}	0.029	0.025
A3	6.26×10^1 – 5.33×10^2	297.80	7.49×10^{-4} – 4.16×10^{-2}	0.021	0.024
A4	3.34×10^1 – 5.38×10^2	285.70	3.69×10^{-4} – 8.31×10^{-2}	0.041	0.049
A5	6.81×10^1 – 5.33×10^2	300.55	7.39×10^{-4} – 4.21×10^{-2}	0.021	0.020
A6	7.11×10^1 – 5.35×10^2	303.05	5.74×10^{-4} – 5.56×10^{-2}	0.028	0.032
A7	6.49×10^1 – 5.35×10^2	299.95	5.30×10^{-4} – 5.83×10^{-2}	0.029	0.033
A8	6.88×10^1 – 5.33×10^2	300.90	7.07×10^{-4} – 4.60×10^{-2}	0.023	0.026
A9	3.95×10^1 – 5.36×10^2	287.45	4.16×10^{-4} – 6.95×10^{-2}	0.034	0.040

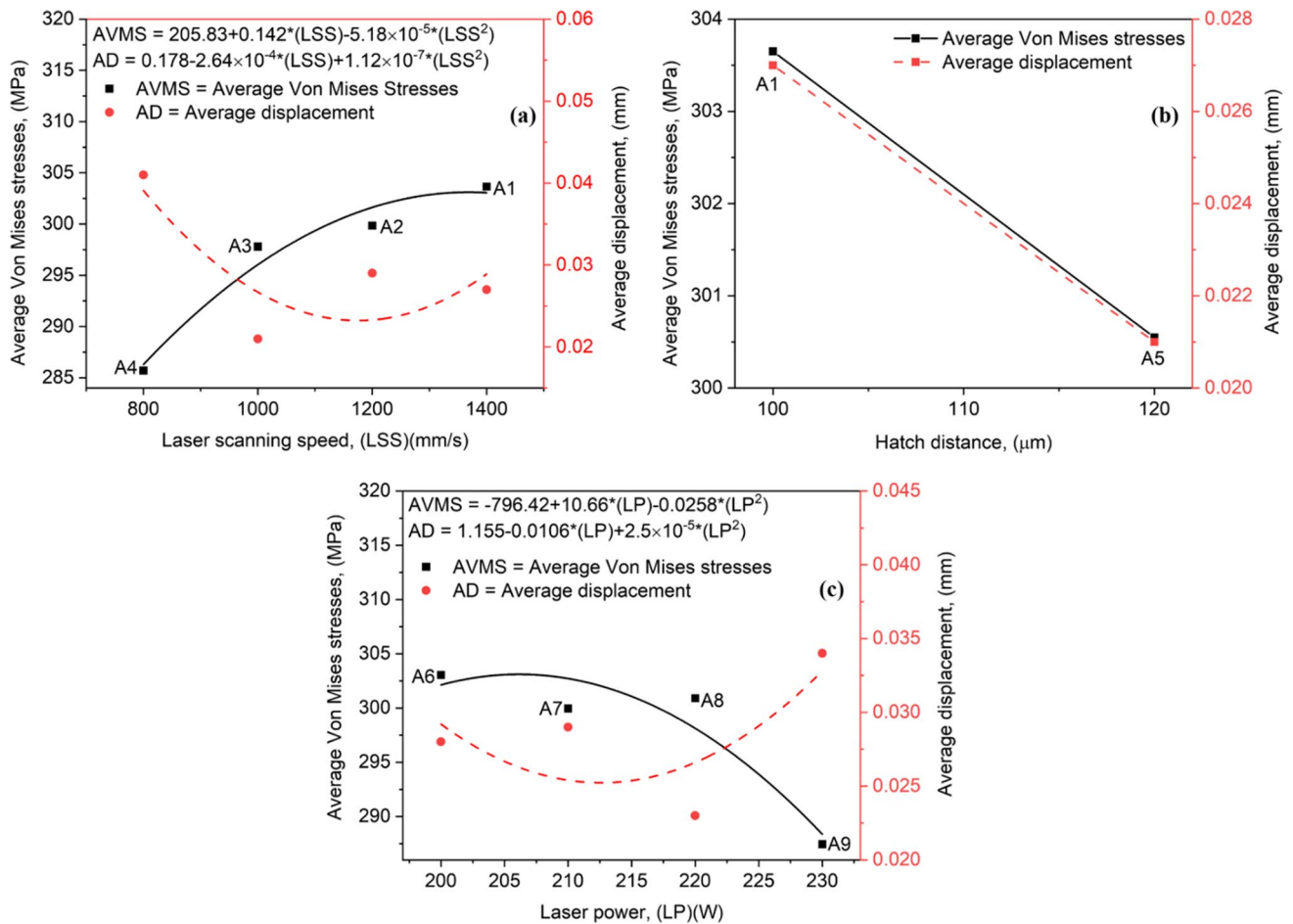


Fig. 8 Effect of **a** laser scanning speed, **b** hatch distance, and **c** laser power on Von Mises stresses and displacement in the LPBF-ed SS 316L parts

correlation among laser power (LP), AVMS, and AD can be analyzed in Fig. 8c. The trend shows that an increase in LP decreases the AVSMS at the verge of AD. Increasing the LP from 200 to 210 W leads to a decrease in AVSM from 304 to 300 MPa. As the LP increases from 210 to 220 W, there is a sudden rise in the AVSM from 300 to 301 MPa. When the LP changes from 220 to 230 W, the AVSM decreases significantly from 301 to 287.5 MPa. In the case of the AD, when the LP increases from 200 to 230 W, the AD elevates from 0.026 to 0.035 mm. To sum up, from data trend analyses, it has been identified that the LSS showed a proportional and an inverse link with AVSM and AD, respectively, while HD presented an inverse relationship with AVSM and AD quantities. Besides, the LP exhibited an inverse relationship with AVSM and a proportional relationship with AD. In LPBF, this behavior can be explained by the laser-material interaction time and the corresponding laser energy transfer inside the material. Keeping the other operating conditions constant, increasing the melt pool cross-sectional area by reducing the LSS/decreasing the HD/increasing LP reduces the cooling rate. It elevates the

material's shrinkage within and around the molten pool, thus reducing the residual stress magnitude [23].

The finite element (FE) simulations were applied for operand monitoring the thermal distribution during the LPBF printing to estimate the average thermal distribution (ATD). Figure 9 compiles a correlation among ATD, Von Mises stress (AVMS), and average displacement (AD). An inverse relationship is calculated between ATD and AVSM, while a proportional correlation is identified between ATD and AD. In LPBF, the primary operating conditions, including laser power (LP), laser scanning speed (LSS), powder layer thickness (PLT), and hatch distance (HD), define the volumetric energy density (VED), which in return controls the regional ATD. VED is defined as [58]

$$VED = \frac{LP}{LSS \times HD \times PLT} \quad (10)$$

From literature [59], a proportional correlation has been identified between VED and ATD. For a given layer, the VED and resultant ATD control the melt pool dimensions

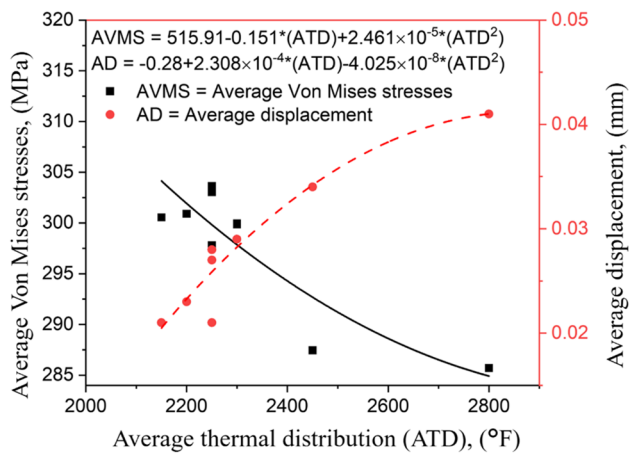


Fig. 9 Correlation among average thermal distribution, average Von Mises stresses, and average displacement in the LPBF-ed SS 316L parts

and cooling rate such that a reduced VED will result in a decreased ATD with a lessened melt pool dimension and cooling rate. It will promote the irradiated material's AD within and around the periphery of the molten pool and reduce the AVSM [23].

3.3 Experimental analyses

Figure 10 discusses the effect of laser scanning speed (LSS) and hatch distance (HD) on the printed samples' morphologies captured via a scanning electron microscope (SEM). Here, the scaling units show the magnification such that $400 \mu\text{m} = \times 260$, $200 \mu\text{m} = \times 500$, $100 \mu\text{m} = \times 1000$, and $20 \mu\text{m} = \times 4000$, respectively. For the A1–A4 samples, LSS decreased from 1400 to 800 mm/s, while in the case of samples A1 and A5, HD increased from 100 to 120 μm . Defects, including un-melt powder particles and lack of fusion, were identified in all the samples.

In LPBF of metallic alloys, the major defects include (a) porosity, (b) lack of fusion, and (c) cracks [60]. The primary factors involving laser power (LP), laser scanning speed (LSS), hatch distance (HD), powder layer thickness (PLT), and printing conditions play essential role in generation. The primary reason behind such defects is an inappropriate selection of processing parameters. The porosity-based defects usually occur when the powder particulates with “hollow structure” are utilized during printing. During LPBF process, the temperature of the localized region inside the melt pool is usually very high owing to the rapid heating with very slow conduction, which increases the “gas” dissolution inside the molten material, promoting porosity. In powder preparation, the gas prepared is usually applied to transform the ingot into micro-granules. During powder manufacturing, the blown gas is entrapped inside

the powder particulates, leading towards porosity formation [61]. The keyhole formation is another important source of porosities in LPBF. In a given liquid, cold zones with elevated surface tension and reduced rebound pressure are the primary cause of keyhole formation [62]. As the keyhole forms, the bubbles float and attempt to escape from the liquid free surface and link with other bubbles and grow in volume. The porosity results when the liquid front solidifies before the bubbles escape from the liquid [63]. Incomplete fusion, referred to as lack-of-fusion (LOF) defects, occurs when the provided energy density (ED) is not sufficient to melt the powder particles completely. LOF can be further divided into inadequate bonding and un-melted regime. In LPBF, when the volume energy density (VED) is very low, the molten pool dimensions reduce and do not provide enough overlap between the two deposited layers, thus providing un-melted patches. When a fresh layer is being deposited on the previous layers, it becomes impossible to re-melt the un-melted particles involved in the last layer. Failures of the molten pool penetration depth can be caused by weak interlayer bonds if VED is insufficient. As a result, laser scan tracks and deposited layers frequently contain LOF defects. A liquid metal flow involved in the molten material enhances the surface roughness, thus causing interlayer defects [64]. An interlayer defect may develop and propagate in a continuous deposition process, resulting in enormous multi-layer errors [65]. LPBF printing often involves the rapid heating, melting, and solidification of metallic powders as a result of localized VED. Due to the 108 K/s cooling rate usually involved in the molten pool, the samples are subjected to thermal stress during printing [66, 67]. Because of the strong heat gradients and high residual stresses in the manufactured parts, cracks begin and spread rapidly in the LPBF-ed parts [66, 67]. As shown in Fig. 10, LOF due to inadequate bonding and un-melted powder particles was identified.

The effect of laser power (LP), from 190 to 200 W, on the printed samples' morphology can be analyzed in Fig. 11. All the images have been produced using a scanning electron microscope (SEM). Here, the achieved morphologies also presented the un-melted powder particles and lack-of-fusion-based defects. The explanation for the LOF-based defects is the same as Fig. 10.

For Figs. 11 and 12, image processing via “ImageJ” software was carried out to determine the porosity in each area using SEM images. For this purpose, images were introduced in the software, and a scale was set using “set scale” command. Following on, the image type “8-bit” was selected, and “threshold” command was applied to highlight the defect region within the SEM image of each sample. The total area of the defective region was measured and the following equation was applied to calculate the porosity in a given area:

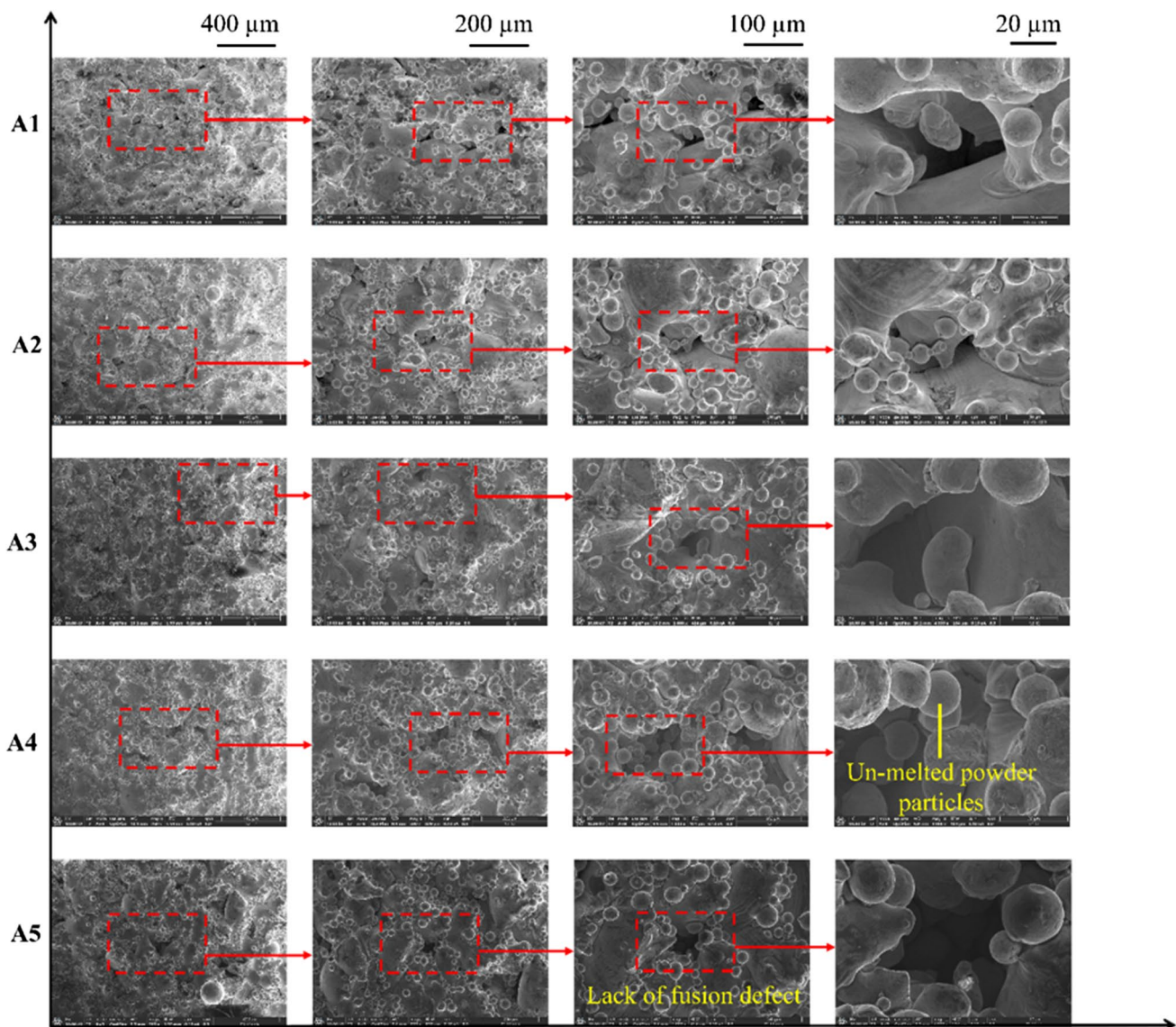


Fig. 10 Effect of laser scanning speed (A1–A4) and hatch distance (A5) on surface morphology of LPBF-ed SS 316L by a scanning electron microscope

$$\text{Porosity in a given area (\%)} = \frac{\text{Defective area}}{\text{Total area}} \times 100 \quad (11)$$

The porosity (%) for each sample is provided in Table 7.

Figure 12 shows the destructive deformation due to the Rockwell hardness indentation test on the SS 316L cube. It is important to mention here that the indentation has been shown for A1 sample only.

Table 8 shows the hardness values for all the A1–A9 samples.

Figure 13 reveals the effect of laser scanning speed (LSS) on phase transformation in the LPBF-ed SS 316L samples (A1–A4). X-ray diffraction (XRD) patterns were indexed using the JCPDS (card number. 33–0397).

From the XRD pattern, only the face cubic centered (FCC) austenite-gamma (AG) phase has been identified at three different orientations, including (111), (200), and (220). Decreasing the LSS from 1400 to 1200 mm/s leads to AG (111) intensity decrease from 150 to 130, while AG (200) intensity increases from 90 to 100; however, almost no effect has been observed for AG (220) intensity (= 58). When LSS further decreases from 1200 to 1000 mm/s, a sudden rise in AG (111) intensity can be observed (130 to 158) at the verge of AG (200) intensity, which decreases from 100 to 98, while AG (220) intensity elevates from 58 to 85. Gradually moving from 1000 to 800 mm/s LSS causes a further declination in AG (111) intensity counts from 158 to 98 and an increase in

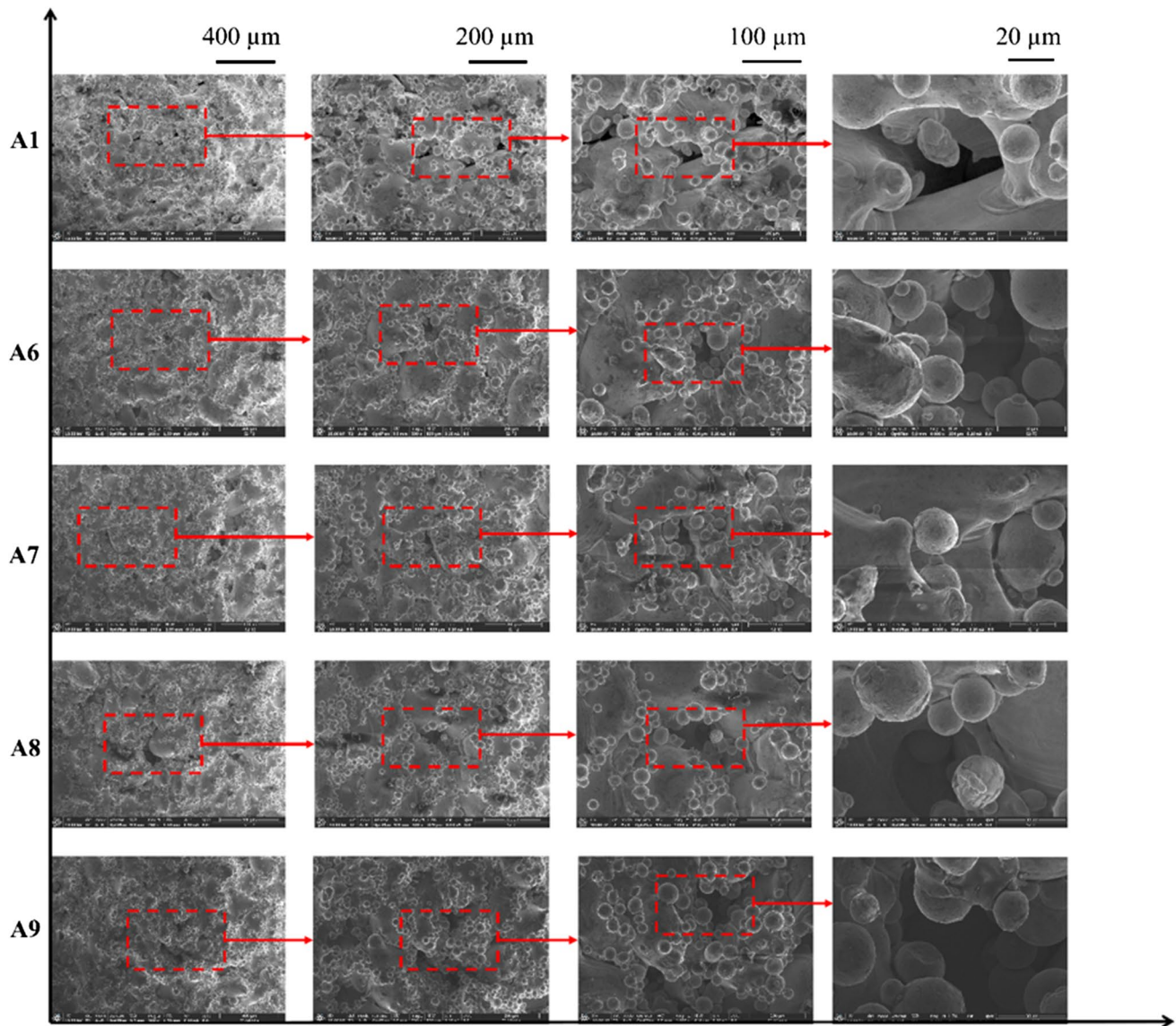


Fig. 11 Effect of laser power on surface morphology of LPBF-ed SS 316L by scanning electron microscopy

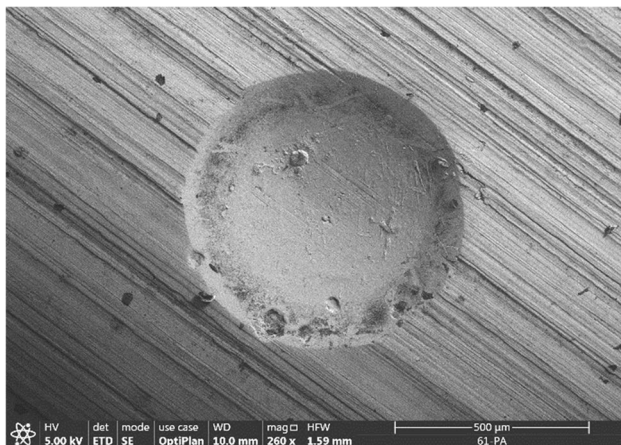


Fig. 12 SEM of Rockwell hardness test indentation on SS 316L

Table 7 Porosity for each sample

Sample No	Porosity (%)
A1	3.99
A2	1.50
A3	3.45
A4	1.32
A5	2.85
A6	3.50
A7	3.60
A8	3.70
A9	3.65

Table 8 Rockwell hardness values for A1–A9 samples

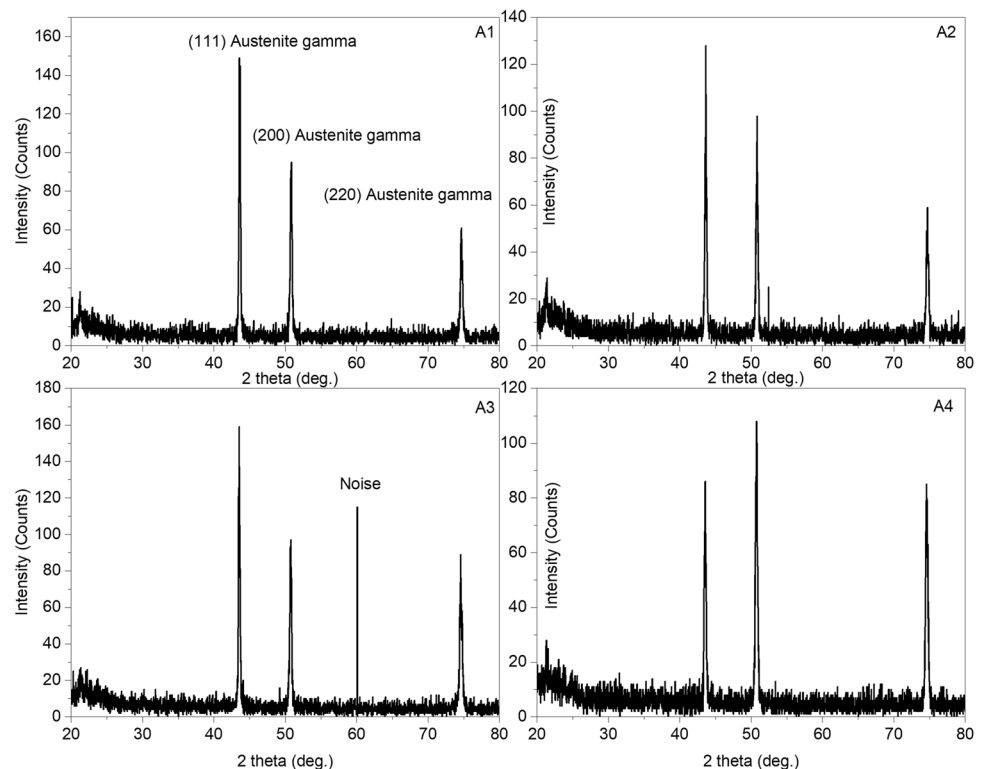
Sample No	Rockwell hardness test value
A1	90
A2	88
A3	87
A4	89
A5	90
A6	91
A7	92
A8	94
A9	95

- The laser beam used in LPBF can fluctuate in intensity or focus, resulting in flaws such as porosity, partial melting, or incomplete layer fusing.
- Due to thermal gradients, temperature fluctuations, and thermal expansion and contraction during cooling, LPBF might cause residual stress in the final product. These residual pressures might cause the final product to break, deform, or distort.
- Inadequate powder distribution can lead to inadequate melting or incomplete fusion between layers, resulting in flaws in the final product.
- Inadequate laser parameters, such as laser power, scanning speed, or hatch spacing, can cause flaws such as porosity, insufficient melting, or inadequate layer fusing.

AG (200) intensity from 98 to 110. At this LSS, almost no change was observed in AG (220) intensity (= 85). In Fig. 13, aberrations can be observed in the XRD pattern that may be contributed to the “impurities” or “defects” in the LPBF parts. In LPBF, impurities or defects can be generated in several ways [68, 69]:

- Contaminated powders can introduce impurities into the LPBF process. This can occur throughout the powder production process, as well as during powder handling, transit, and storage. Moisture, oxygen, and other gases, as well as foreign particles, can all have an impact on the final product's quality.

Impurities in a sample might impact its XRD pattern. XRD is a technique for determining a material's crystal structure by studying the diffraction pattern created by X-rays interacting with the material's atomic structure. The pattern is determined by the material's crystal structure, and any impurities or faults in the crystal structure might impact the diffraction pattern. Impurities or flaws can create new crystallographic planes or distort existing ones, leading in more peaks or shifts in peak locations in the XRD pattern. These variations can make identifying the crystal structure of the sample challenging, and the presence of impurities or flaws can also impact the strength of the peaks in the XRD pattern.

Fig. 13 Effect of laser scanning speed on X-ray diffraction pattern during LPBF of SS 316L

An effect of hatch distance (HD) on phase transformation can be observed in the LPBF-ed SS 316L samples using Fig. 14. Here, only austenite-gamma (AG) has been identified at three different orientations, including (111), (200), and (220). As HD increased from 100 to 120 μm (A1 and A6), the AG (111), AG (200), and AG (220) phase intensities decreased from 150 to 128, 90 to 77, and 58 to 52, respectively.

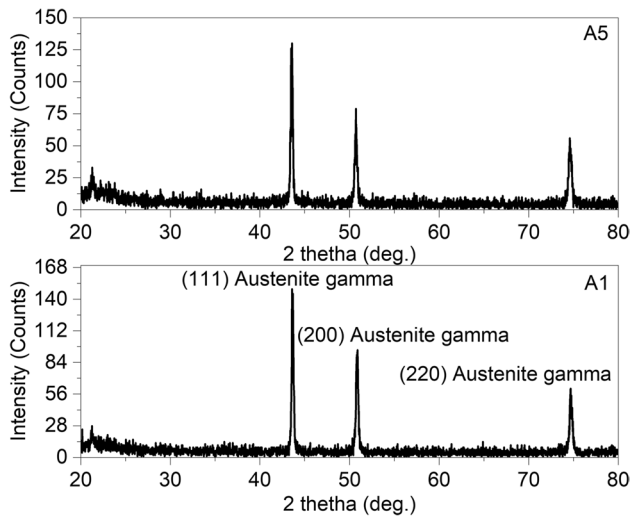


Fig. 14 Effect of hatch distance on X-ray diffraction pattern during LPBF of SS 316L

Figure 15 exhibits the influence of the laser power (LP) on the phase transformation in the LPBF-ed SS 316L parts. Sample A1 has been used as a reference to compare the results with the rest of the specimens (A6–A9). In all the printed samples, austenite-gamma (AG) phase has been identified with various orientations such as (111), (200), and (220). When LP increases from 190 to 200 W, AG (111) and AG (220) intensities elevate from 150 to 160 and 58 to 70, respectively, while AG (200) intensity declines from 90 to 80. With the further increment in LP from 200 to 210 W, an opposite effect has been identified on AG (111) intensity such that its value decreases from 160 to 130 while AG (200) intensity increases from 80 to 130. However, AG (220) intensity counts remain near to the constant ($= 71$) at 210 W LP. Replacing the LP equal to 210 with 220 W causes an increment in AG (111) and AG (200) intensities from 130 to 175 and 130 to 180, respectively, at the cost of AG (220) intensity count decreased from 71 to 60. This trend was again disturbed when the LP increased further from 220 to 230 W, resulting in AG (111) intensity equal to 110 (decreased from 175), AG (220) intensity equal to 138 (decreased from 180), and AG (222) remained near to constant ($= 61$). Hence, based on the provided experimental analyses, it can be summarized that the LP presented a random effect on the AG (111), AG (200), and AG (220) intensities during the LPBF of SS 316L.

Figure 16a compiles the effect of laser scanning speed (LSS) on the average surface roughness (R_a), average ten-point height (R_z), and RMS roughness (R_q) along the

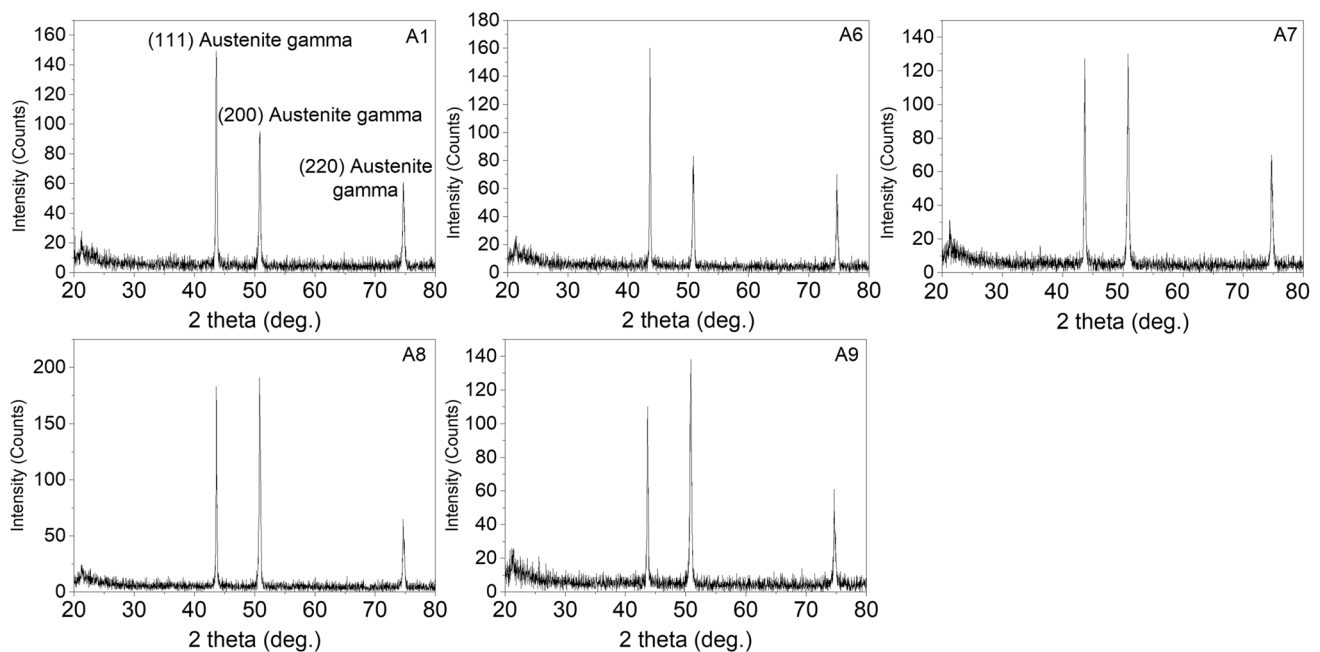


Fig. 15 Effect of laser power on X-ray diffraction pattern during LPBF of SS 316L

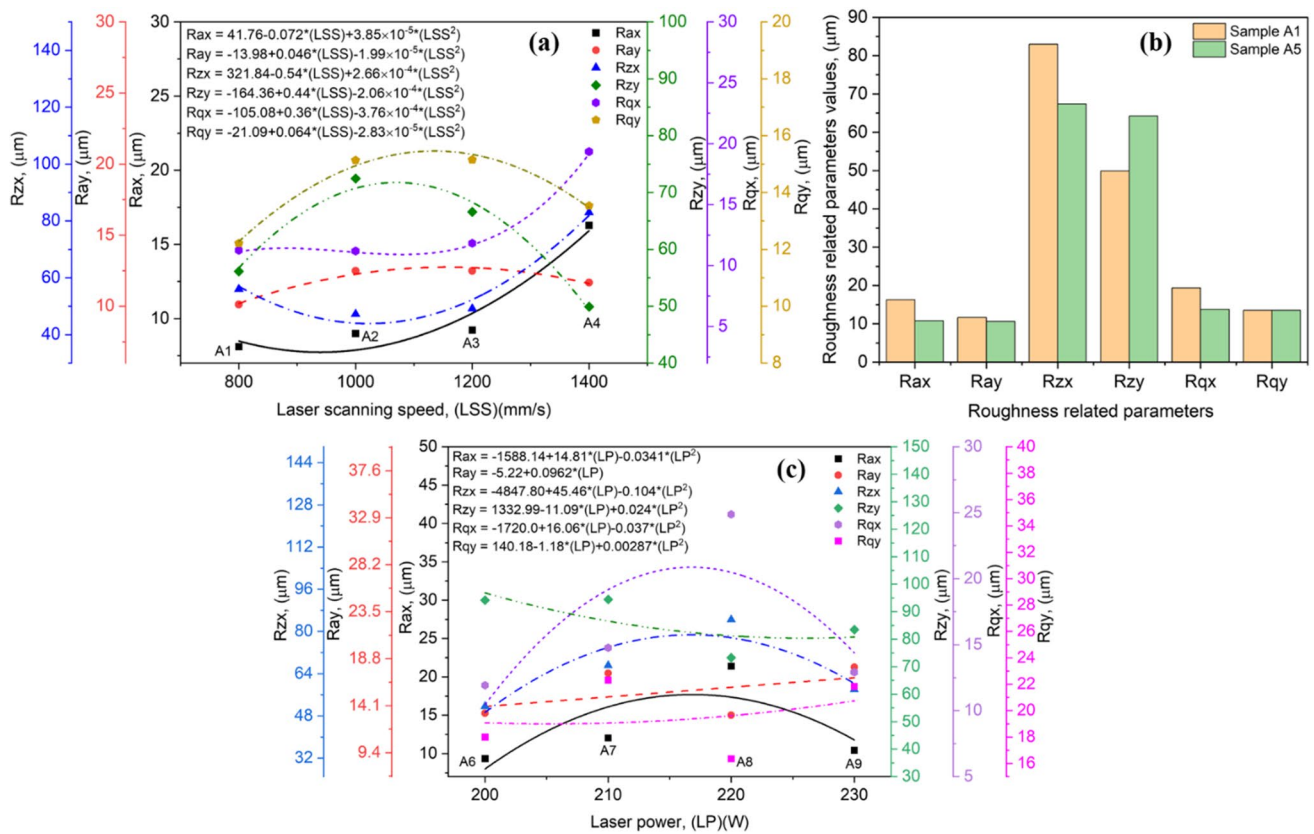


Fig. 16 Correlation among **a** laser scanning speed, **b** hatch distance, and **c** laser power and roughness parameters after LPBF of SS 316L

x - and y -directions. Here, x - and y -directions are parallel and perpendicular to the powder spreader movement (PSM). Various studies have been carried out that claimed that the roughness parameters were found with a deviation in both directions [46, 47, 70, 71]; this is why parameters have been calculated in two-defined orientations. For Ra, Rz, and Rq, the data trend shows that with an increment in LSS, the Rax, Rzx, and Rqx increase significantly, while an opposite trend has been presented with Ray, Rzy, and Rqy. When the LSS is low, the high volumetric energy density (VED) causes the powders to melt entirely, achieving a flat surface along the x -direction. Usually, as the LSS grows, the tracks may convert from continuous to irregular shape, suggesting that an unstable condition has been approached. In other words, the Plateau-Rayleigh (P-R) unstable state has contributed to such a condition owing to the high LSS [47]. The Plateau-Rayleigh instability contributes to this unstable condition when a considerably high scan speed is applied in LPBF. Various authors have expressed that the molten layer can be classified as the “cylinder” in the liquid form, having a length (L) and diameter (D), resulting when the laser beam falls on the powder bed [72]. A deposited track becomes unstable when the ratio between L/D is higher than “Pi.” In another study [73], it has been identified that the stability of

the deposited track depends on the contact angle between the molten layer and the base plate, showing that the layer will be stable when the contact angle $< \pi/2$. However, this condition is nearly impossible to sustain when a new layer is deposited on the previously printed layer. The molten layer swelling and shrinking promote the “necking,” thus introducing distortion in the deposited layer. Due to this phenomenon, the molten breaks into mini droplets to lower the surface energy, designated as balling. The deposited molten layer break time (BT) can be calculated as [74]

$$BT = \left[0.3433 \sqrt{\frac{\gamma_{LV}}{\rho \left(\frac{D}{2}\right)^3}} \right]^{-1} \quad (12)$$

Here, γ_{LV} and ρ are the material’s surface tension and density, respectively. If the BT is below the solidification time, the probability of balling phenomenon is significant [75]. Besides, with an increment in LSS, the melt pool dimension in terms of length increases, elevating the L/D ratio, promoting the P-R instability and corresponding roughness parameter along the x -direction.

Figure 16b shows the correlation between roughness parameters and hatch distance (HD). It can be seen that

with an increment in HD from 100 to 120 μm , the Rax, Ray, Rzx, and Rqx decrease from 15 to 10 μm , 11 to 10 μm , 85 to 69 μm , and 20 to 15 μm while Rzy increases from 50 to 65 μm . However, the Rqy remains constant ($= 10 \mu\text{m}$) with the elevation of HD. Usually, when the HD increases, the gap between two tracks also increases, which results in a large height difference peaks and valleys of the tracks, resulting in a high roughness [70]. As the HD decreases, the overlap between the two adjacent tracks improves significantly, resulting in a relatively flat surface. This behavior can be explained by the “humping” phenomenon using Marangoni flow (MF) when a reduced HD is usually applied. The Marangoni convection usually occurs when the molten material flows from the low surface tension region towards the higher one within the molten pool [76]. Note that surface tension and thermal distribution are inversely proportional to each other. The thermal distribution value is usually higher in the molten pool central region than the periphery; hence, the MF direction is from the central area towards the edges, transferring mass from the current molten layer to the prior printed layers. The MF usually transfers the molten material to fill up the gaps between the printed layers when the HD is comparatively high [70]. The thermal conductivity also plays an essential role in enhancing the MF for a given material such that a higher thermal conductivity increases the thermal gradient and corresponding MF [77]. At low HD, the thermal gradient increases due to the heat accumulation resulting in a high temperature inside the molten pool, reducing the molten material’s viscosity which is beneficial for MF [78]. This phenomenon increases the MF flow rigorously and the mass transfer intensity to the prior printed layers, tempting the material to stack up and hump as the material is solidified.

Figure 16c reveals the correlation between laser power (LP) and roughness parameters. From the provided data trend, one can observe that the Rax, Rzx, Rzy, and Rqx presented an inverse correlation with the LP, while a slight proportional correlation can be analyzed among LP, Ray, and Rqy. At low LP, the deposited layers do not overlap effectively, which induces a considerable gap between the tracks and corresponding poor quality of the printed regional surface. However, at a relatively high LP, the deposited tracks overlap effectively, eliminating the gap between the tracks, and the surface flatness elevates, increasing the surface quality. Sometimes, roughness increases with an increment in LP due to the balling-based defect in the LPBF, which can be attributed to the reverse MF. As described above, the MF is from the molten pool center towards the edges, which induces the splashing of molten material in the neighboring deposited layers. For a given material, a few elements having an elevated surface activity may lessen the molten material’s surface tension, which can alter the surface tension and MF direction from the periphery towards the center of the melt

pool. Here, a high quantity of molten material accumulates, resulting in material agglomeration, similar to balling, thus affecting the surface morphological quality negatively. In one of the studies, it has been determined that the high LP increases the thermal gradient that may introduce a high surface tension gradient and MF along with the part-build direction [79], increasing the defect probability at high LP. Hence, this balling type is more likely to occur at high laser power.

3.4 Correlation between simulations and experimental parameters

After carrying out the simulation and experimental analyses, a correlation was developed between average thermal distribution (ATD) and austenite-gamma (AG) phase intensity at different orientations, including (111), (200), and (220), as shown in Fig. 17. ATD is estimated using simulations, while the AG-phase intensity has been calculated using experiments (X-ray diffractometry). The data trend shows that ATD has presented an inverse relationship with AG (111), while a proportional correlation has been identified between ATD and AG (220). The trend identified between ATD and AG (200) exhibits a non-consistent nature between the two items in such a way that AG (200) intensity increases from 2150 to 2450 $^{\circ}\text{F}$ and decreases afterward. ATD is controlled by volume energy density in LPBF, which regulates thermal gradient, solidification rate, microstructure, phase intensity, and stresses [80].

Figure 18a–c shows the effect of average thermal distribution (ATD) on roughness parameters, including average surface roughness (Ra), average ten-point height (Rz), and RMS roughness (Rq) parallel (x) and perpendicular (y) to the powder spreader movement. It can be analyzed that the ATD showed an inverse effect on Rax identified using data trend; however, the Ray value increased in the beginning, up to 2450 $^{\circ}\text{C}$, and declined significantly afterward. The same data trend can be observed in the case of Rzx, Rzy, Rqx, and Rqy in connection with ATD defined by the volume energy density. As the ATD increases, the gap between the adjacent deposited tracks decreases, which smoothens the given surface, thus lessening the roughness parameters.

In Fig. 19, a relationship has been developed between average Von Mises stresses (AVMS) and austenite-gamma (AG) phase detected in three different orientations (111), (200), and (220). From the data trend, AG (111) intensity increases with an enhancement in AVMS, while AG (200) and AG (220) intensities decrease with an increment in AVMS. It can be concluded that the operating conditions, including laser scanning speed (LSS), hatch distance (HD), and laser power (LP), play a critical role in defining the AVMS and the relevant phase intensity. From Fig. 18, it can be analyzed that an increase in LSS or a decrease in HD or LP causes an increment in AVMS, which increases AG (111) intensity and decreases AG (200) and AG (220) intensities.

Fig. 17 Effect of average thermal distribution on the LPBF-ed SS 316L austenite-gamma phase intensity counts

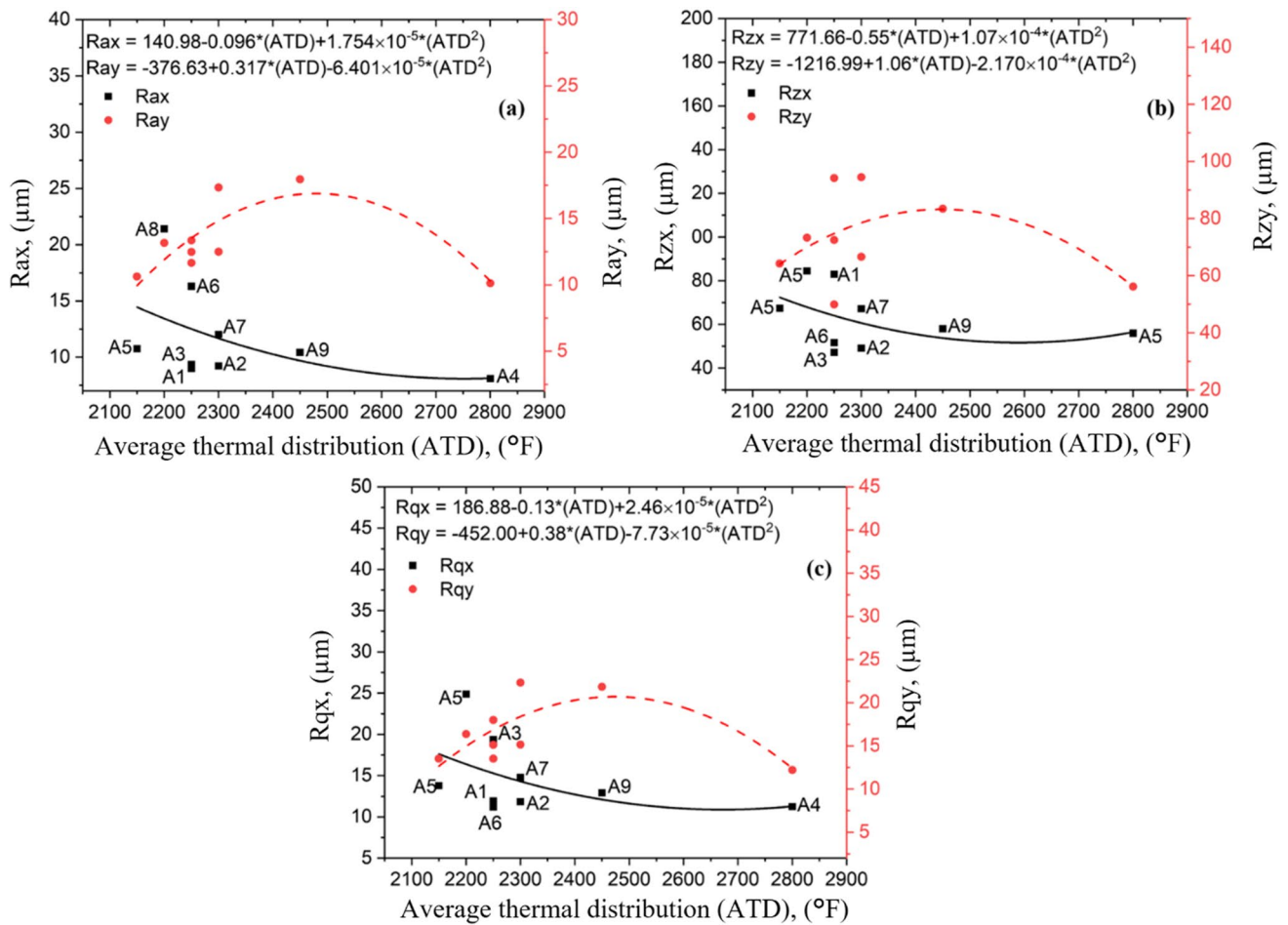
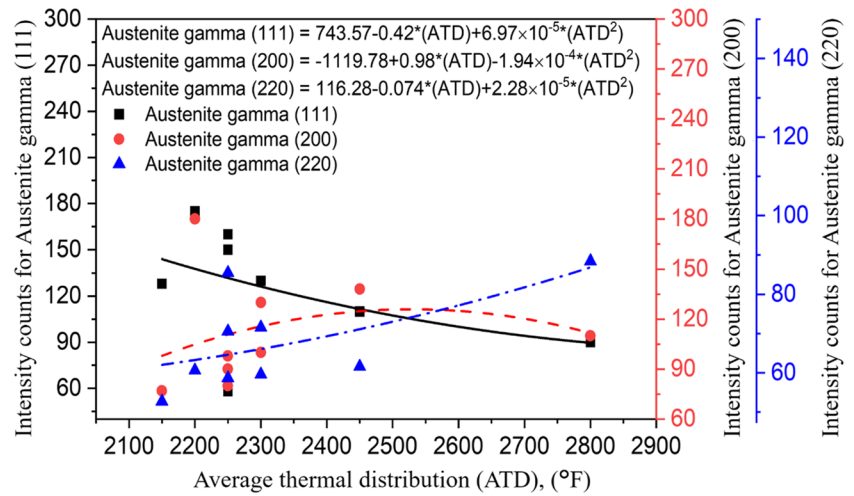


Fig. 18 Effect of average thermal distribution on the LPBF-ed SS 316L roughness parameters **a** average surface roughness, **b** average ten-point height, and **c** average RMS roughness

Fig. 19 Correlation between Von Mises stresses and austenite-gamma phase intensity in LPBF-ed SS 316L specimens

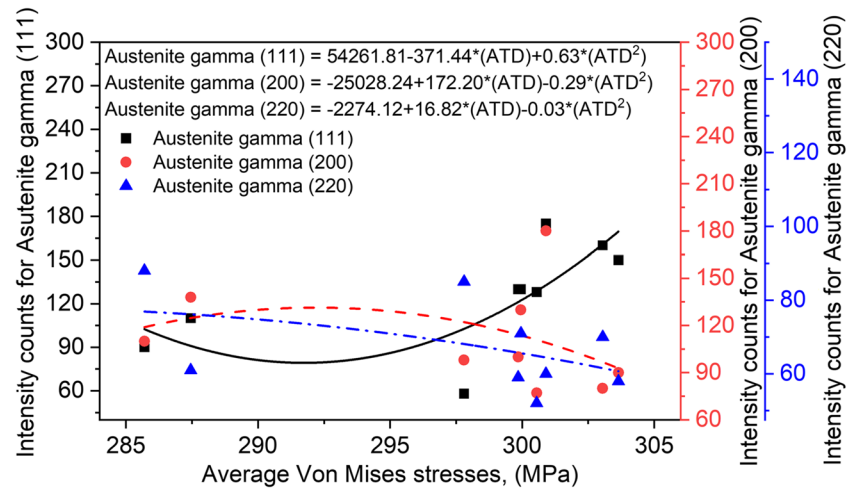


Table 9 Methods to mitigate residual stresses and part deformation in LPBF

Category	Illustration	Reference
Control using process		
Operating parameters	The operational parameters, particularly laser power and scan speed, have a considerable influence on the material's thermal behavior throughout the process. It is feasible to decrease the amount of heat stress and deformation in the part by carefully managing these factors	[81, 82]
Pre- or post-heating	Thermal stress and distortion can be reduced by pre-heating the printing plate or post-heating the final product. Pre-heating the build plate can assist to smooth out the temperature, while post-heating can aid to reduce residual stresses in the part	[83, 84]
Support structures	Support structures can be utilized to assist reduce distortion during the AM process by providing additional support for the component. This is especially handy for complicated geometries and huge pieces	[85, 86]
Optimize the part design	The amount of thermal stress and distortion that happens during the AM process can be significantly influenced by the part's design. It is feasible to lessen the risk of these difficulties arising by optimizing the design	[87, 88]
Control using technologies		
In situ monitoring	In situ monitoring systems employ sensors to measure factors like temperature, pressure, and displacement during the AM process. These sensors can give the control system real-time input, allowing it to change process settings to decrease heat stress and distortion	[89, 90]
Closed-loop control	Closed-loop control systems modify process parameters in real time based on sensor data, allowing for more accurate control of the AM process. Thermal stress and deformation in the completed product can be reduced using closed-loop system	[91, 92]
Thermal simulation software	The temperature behavior of the material during the AM process may be modeled using simulation software. It is feasible to identify potential faults and alter process parameters to decrease thermal stress and distortion in the completed product by modeling the process	[93–95]
Machine learning	To anticipate and manage the thermal behavior of the material during the AM process, machine learning may be trained on data from past AM results. By allowing the control system to make more educated judgments, this can assist to decrease thermal stress and distortion in the completed item	[96, 97]
Control using equipment		
Nanotreating	A thin coating of nanoparticles is applied to the surface of the material being printed during the nanotreatment process. This coating can increase the thermal conductivity of the material, allowing for more equal heating and cooling throughout the AM process. This may decrease heat stress and deformation in the final product	[98–100]
Ultrasonics	The use of high-frequency sound waves to change the material being printed is known as ultrasonic therapy. This can potentially reduce thermal stress and distortion by helping to even out the temperature of the material and relieve residual stresses	[101, 102]
Multi-source treatment	Rather than depending exclusively on a laser, multi-source treatment includes the use of various energy sources to heat and melt the material being printed. This has the ability to minimize thermal stress and distortion by allowing the material to be heated and cooled more evenly	[103, 104]

4 Conclusions

This study proposes a 3D transient thermo-mechanical finite element (FE) simulation model with temperature-dependent properties and experimentally validated for the part-scale modeling of laser powder bed fusion (LPBF) of SS 316L. The effect of operating conditions, including laser scanning speed (LSS), hatch distance (HD), and laser power (LP), was analyzed on the melt pool evolution, residual stresses induction, phase evolution, and morphology as a function of the number of layers. The significant findings are summarized here.

- Increasing the number of layers causes an increment in thermal distribution and melt pool dimensions (depth, width, and length) due to cyclic heating and cooling cycles in laser-based printing processes. The melt pool size increases by either increasing the LP or decreasing the HD or LSS, which will raise the volume energy density transfer to the particular region.
- Compressive and tensile stresses have been monitored in the printed parts. The Cauchy stresses, along xx- and yy-orientations, are maximum at the bottom and top surfaces of the printed part. Along the zz-orientation, these stresses are higher than the xx- and yy-orientations in the whole printed cube except the top surface. Von Mises stresses are minimal compared to the Cauchy stresses.
- An inverse correlation has been identified for average Von Mises stresses (AVMS) with LP and AVMS with HD, while a proportional relationship is presented between LSS and AVMS. On the other hand, the average displacement (AD) showed an inverse correlation with LSS and HD; however, a proportional relationship between LP and AD is identified. Average thermal distribution (ATD), controlled by the volume energy density, exhibited an inverse effect on AVMS and a proportional effect on AD.
- The printed samples' morphologies showed a lack-of-fusion defects. The X-ray diffractometry exhibited austenite-gamma (AG) phase in the LPBF-ed parts in three different orientations (111), (200), and (220) with the intensities dependent on the operating conditions. A decrease in either HD or LP decreased the average surface roughness (AVS) while an increase in LSS causes an increment in AVS in a direction parallel to the powder spreader movement (x). However, the AVS perpendicular to the powder spreader movement (y) showed an inconsistent behavior for LSS, HD, and LP.
- ATD induced a negative impact on AG (111) and AG (200) intensities and a positive influence on AG (220). Furthermore, an increment in ATD leads to a decrement in roughness parameters, including AVS, ten-

point height, and RMS roughness, along the x -direction. In contrast, an opposite trend was identified along the y -direction. The AVMS revealed a proportional link with AG (111) intensity and an inverse correlation with AG (200) and AG (220) intensities count.

This study developed a detailed correlation among operating conditions melt pool evolution, residual stresses, phase formation, morphology, and number of layers, in the case of LPBF of SS 316L.

5 Future outlook: methods to mitigate residual stresses and part deformation

Table 9 discusses the methods to mitigate the residual stresses and part deformation during LPBF process.

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Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Competing interests The authors declare no conflict of interest.

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